國立陽明交通大學

智慧系統與應用研究所

碩士論文

Institute of Intelligent Systems

National Yang Ming Chiao Tung University

Master Thesis

FieldAnchor：農田無人機—衛星資料集與無 GPS 導航局部視覺重定位之硬式均值漂移解碼

FieldAnchor: A Farmland UAV–Satellite Dataset and Hard Mean-Shift Decoding for GPS-less Navigation

研究生：李宇弘（Yu-Hong Lee）

指導教授：謝君偉（Jun-Wei Hsieh）

中華民國一一五年八月

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A Thesis
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in partial Fulfillment of the Requirements
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in
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致謝

兩年的研究所生涯即將告一段落，回顧這段求學歷程，首先要誠摯感謝我的指導教授 謝君偉教授。在研究進行的過程中，教授持續透過定期會議掌握研究進度，並針對研究問題、實驗設計與論文撰寫給予許多寶貴的建議。當研究遭遇瓶頸或實驗結果不如預期時，教授也會引導我重新檢視問題，從研究動機、方法設計到實驗設定逐步分析，協助我釐清問題的核心。透過這段時間的訓練，我逐漸學會在面對複雜問題時，先將問題拆解，再依序驗證可能的原因與解決方案，而不是急於尋求答案。這樣的思考方式不僅運用於研究與程式開發，也使我在處理各種挑戰時更加有條理。感謝教授一路以來的指導與督促，使我能順利完成研究，也培養了獨立思考、解決問題及面對壓力的能力。

此外，也要感謝研究室的所有夥伴，讓忙碌的研究生活增添了許多溫暖與值得珍藏的回憶。感謝東琳同學在我剛進入研究室、尚未熟悉環境時，熱心分享相關經驗並提供協助，使我能更快適應研究所的生活。感謝晉翔與博云在論文進入最後階段時，陪伴我度過許多長時間留在實驗室的日子。無論是一起用餐、討論研究，或是在疲憊時到戶外走走、聊聊近況，這些看似平凡的日常，都成為支撐我繼續努力的重要力量。同時，也感謝研究室其他同學平時的關心與陪伴，大家一起聊天、運動、聚餐及玩桌遊的時光，讓我能在緊湊的研究步調中獲得短暫的放鬆。因為有各位的陪伴，這段研究所生活不只有實驗與論文，也留下了許多真誠而美好的回憶。

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學生：李宇弘　　指導教授：謝君偉 教授

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摘要

全球定位系統（Global Positioning System, GPS）是無人機取得絕對位置的重要來源；當 GPS 受干擾或遭拒止時，無人機需藉由視覺資訊將當前影像對應至地理參考衛星圖。農田具有重複田區、平行田埂與相似植被紋理，使鄰近候選容易產生混淆。本文提出 FieldAnchor 無人機—衛星資料集，並提出 FieldAnchor Local Relocalization（FieldAnchor-LR）局部視覺重定位框架，在已知預定航線與受控粗略位置條件下進行定位。預定航線由起點、依序定義的參考點與終點構成；每次定位先以粗略位置建立完整 6×6（36 個）局部衛星候選幾何，再依因果前向方向進行硬式候選限制，只保留前方 3×6（18 個）候選，後方 18 個候選在在線投影與相似度計算前即排除。保留候選經凍結的 MobileCLIP2-S2 與雙分支投影器取得跨視角特徵，接著以均值漂移（Mean-Shift, MS）整合空間響應並估計連續視覺位置與不確定度。門控循環單元（Gated Recurrent Unit, GRU）融合最近三幀影像特徵、衛星脈絡及前一時刻狀態，估計速度、加速度與量測資訊；二階慣性多項式形成下一步運動預測，最後由不確定度感知卡爾曼濾波（Kalman Filter, KF）融合運動預測與視覺量測，得到連續定位結果。本文標題中的「Hard」僅指 Mean-Shift 解碼前由完整 6×6 幾何硬式保留前向 3×6 候選的限制，並非另一種獨立的 Mean-Shift 演算法。

關鍵字： 無人機定位、衛星影像、跨視角地理定位、均值漂移、時序定位、門控循環單元、卡爾曼濾波。

FieldAnchor: A Farmland UAV–Satellite Dataset and Hard Mean-Shift Decoding for GPS-less Navigation

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Abstract.

Global Positioning System (GPS) measurements are an important source of absolute position for unmanned aerial vehicles (UAVs). When GPS is disrupted or denied, visual observations can be matched to a georeferenced satellite map to recover position. Repetitive field boundaries and similar vegetation make this difficult in farmland because nearby satellite candidates can appear alike. This thesis introduces FieldAnchor, a UAV–satellite dataset, and proposes the FieldAnchor Local Relocalization (FieldAnchor-LR) framework under a known planned route and a controlled coarse-position condition. The planned route is represented by a start point, ordered predefined reference points, and a destination. For each query, a complete 6×6 local satellite geometry containing 36 candidates is first established around the coarse position. A hard forward candidate restriction is then applied before online visual scoring: only the route-aligned forward 3×6 subset containing 18 candidates is retained, while the rear 18 candidates are excluded before satellite projection and similarity computation. A frozen MobileCLIP2-S2 encoder with view-specific projectors forms cross-view features for the retained candidates, and Mean-Shift (MS) aggregates the spatial response to estimate a continuous visual position and uncertainty. A three-frame Gated Recurrent Unit (GRU) combines recent image features, satellite context, and the previous state to estimate motion and measurement information. A second-order inertial polynomial predicts the next displacement, and an uncertainty-aware Kalman Filter (KF) fuses the motion prediction with the visual measurement to obtain the final continuous position. The term “Hard” refers only to the explicit forward candidate restriction applied before Mean-Shift decoding; it does not denote a separate variant of the Mean-Shift algorithm.

Keywords: UAV localization, satellite imagery, cross-view geo-localization, Mean-Shift, temp oral localization, Gated Recurrent Unit, Kalman filtering.

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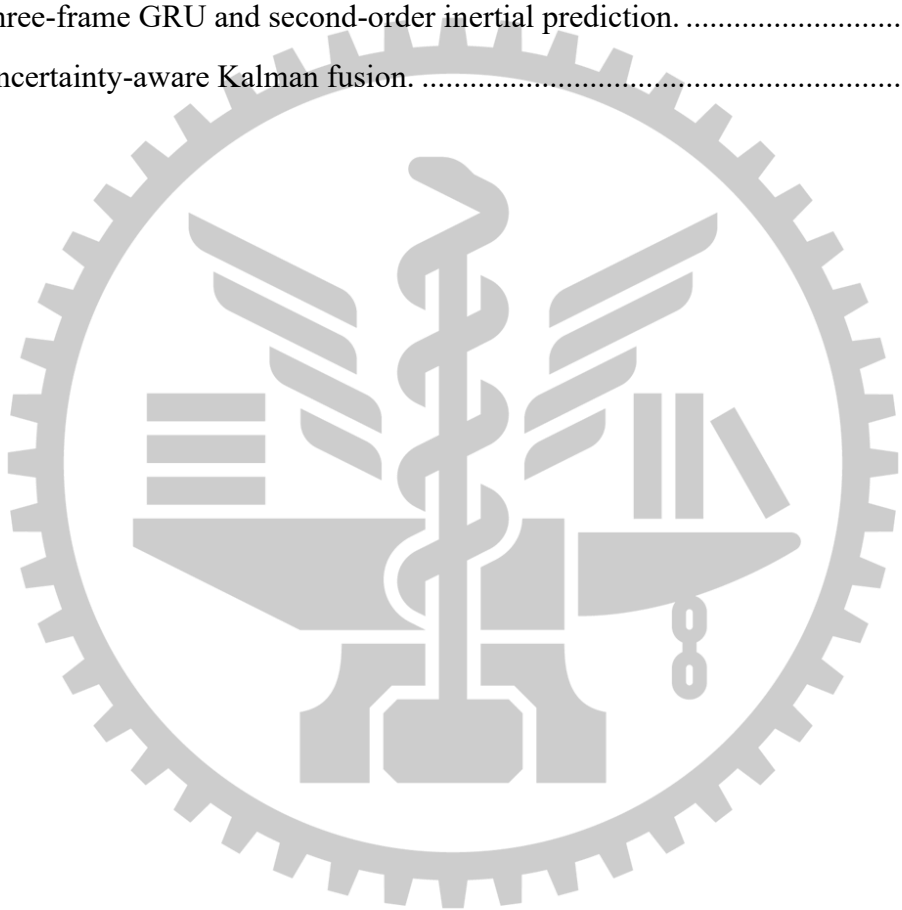

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Chapter 1 Introduction

1.1 Research Background and Motivation

Global Navigation Satellite System (GNSS) measurements are a common source of absolute position for an unmanned aerial vehicle (UAV), but their availability and accuracy can deteriorate under signal blockage, interference, or denied environments. A complementary visual localization module can recover position information by matching the current UAV observation to a georeferenced satellite or orthophoto map. The problem is particularly difficult in agricultural areas, where repeated field boundaries, irrigation channels, access roads, and similar crop textures can generate several visually plausible satellite responses within a compact neighborhood.

Cross-view geo-localization has progressed from global image retrieval toward UAV self-positioning and metric localization. Retrieval-oriented methods such as CVM-Net [1], SAFA [2], TransGeo [3], and Sample4Geo [4] learn cross-view representations, while UAV-oriented benchmarks and systems including University-1652 [5], SUES-200 [6], DenseUAV [7], Game4Loc [8], Bearing-UAV [9], and InfoGeo [10] address aerial-to-satellite matching under increasingly realistic conditions. Recent 2026 work further studies noisy correspondence, unified drone–satellite–ground matching, multi-view foundation models, geometry-guided geo-localization, and direct UAV ego-localization [16]–[20]. These advances motivate a local navigation setting in which visual evidence, route structure, temporal motion, and an explicit state estimator are combined rather than treated as isolated independent-frame decisions.

1.2 Problem Definition

This thesis studies route-conditioned local visual relocalization rather than unconstrained global map search. The mission is assumed to provide a known planned route defined by the start point, ordered predefined route reference points, and destination. Each localization query is associated with one reference point on this planned route and a controlled coarse local position estimate that is used only to establish a compact satellite candidate neighborhood. The predefined reference point is retained separately as a supervision and evaluation target; it is not concatenated with the UAV image embedding, satellite visual embedding, or recurrent feature vector, and it is not reported as the localization output. The thesis therefore evaluates visual refine

ment around a known planned route while keeping the route reference annotations separate fro m the learned visual and recurrent representations.

Under this controlled local-search condition, the objective is to refine an imperfect coarse location using UAV–satellite visual evidence while preserving temporal consistency. The repo rted experiments therefore evaluate local visual relocalization along a prescribed route, includi ng metric accuracy, temporal stability, robustness to ambiguous satellite responses, and the co mputational trade-off of forward candidate restriction.

1.3 Research Contributions

The main contributions of this thesis are summarized as follows: (1) FieldAnchor, a real-f light farmland UAV–satellite dataset with a registered orthomosaic and route-held-out evaluati on split; (2) a controlled local-search benchmark based on predefined planned-route reference points, together with a hard forward candidate restriction that reduces a complete 6×6 local ge ometry from 36 candidates to the route-aligned forward 3×6 subset of 18 candidates before on line projection and similarity scoring; (3) a Mean-Shift (MS) decoder that converts the retaine d candidate responses into a continuous visual position and uncertainty; and (4) a three-frame Gated Recurrent Unit (GRU), second-order inertial predictor, learned measurement variance, a nd external Kalman Filter (KF) that improve temporal state estimation while keeping the refer ence points used for supervision and evaluation separate from the model input representation.

Chapter 2 Related Work

2.1 Cross-View Geo-Localization

Cross-view geo-localization determines geographic correspondence between observation s acquired from substantially different viewpoints. CVM-Net [1] and SAFA [2] established lea rned cross-view feature matching and spatial aggregation, while TransGeo [3] introduced trans former-based representations for global geo-localization. Sample4Geo [4] demonstrated the va lue of geographically and visually hard negatives. More recent studies broaden this formulatio n: PAUL [16] targets noisy cross-view correspondence, UniGeoRS [17] unifies satellite, drone, and ground-view localization in one benchmark, GeoBridge [18] incorporates semantic ancho rs across multiple views and modalities, and Geo2 [20] introduces geometry-guided cross-vie w localization. These methods strengthen cross-view representation learning, but they primaril y emphasize retrieval or single-observation geo-localization rather than the causal temporal sta te required by the route-conditioned estimator studied in this thesis.

2.2 UAV–Satellite Localization

UAV-specific datasets and methods have progressively reduced the gap between image re trieval and practical self-positioning. University-1652 [5] provides a multi-view drone benchm ark, SUES-200 [6] studies cross-view changes across scene types and flight heights, DenseUA V [7] targets low-altitude UAV self-positioning with dense sampling and distance-aware evalu ation, and Game4Loc [8] extends UAV geo-localization to continuous areas with metric locali zation error. In 2026, Bearing-UAV [9] directly predicts absolute position and heading from ne ighboring aerial–satellite features, InfoGeo [10] improves cross-view generalization through o bject-centric information modeling, and PiLoT [19] performs UAV ego- and target geo-localiz ation by registering live video against a georeferenced 3D map. These studies motivate the tra nsition from image-level retrieval toward navigation-oriented localization, while FieldAnchor-LR focuses specifically on route-conditioned local UAV–satellite relocalization with causal te mporal state propagation.

2.3 Spatial Response Decoding

Mean shift is a classical non-parametric mode-seeking procedure that iteratively moves a point toward a locally dense region under a kernel-weighted distribution [11]. Hsieh et al. [15]

further demonstrate a mean-shift-based optimization mechanism in differentiable architecture search, where iterative shifting within a selected bandwidth is used to smooth the search landscape and improve stability. Although that work addresses neural architecture search rather than UAV localization, it provides a recent example of using mean shift as a differentiable iterative refinement mechanism. In FieldAnchor-LR, Mean-Shift (MS) is instead applied to georeferenced satellite candidate centers, using the visual posterior as response mass to obtain a continuous visual position. The final implementation does not snap the result to a discrete satellite anchor; shifted modes contribute through density-based weights, and the same response distribution is used to derive visual uncertainty before temporal fusion.

2.4 Temporal State Estimation and Filtering

Temporal localization requires a state representation that can combine current visual evidence with recent motion. The Gated Recurrent Unit (GRU), introduced as a recurrent gating mechanism for sequence modeling [13], provides a compact way to preserve relevant history without using a long explicit image stack. In parallel, Kalman filtering provides a principled recursive framework for combining a motion prior and a noisy measurement [14]. The proposed system uses a three-frame GRU to estimate velocity, acceleration, heading residual, visual correction, and measurement variance, and then applies an external KF whose measurement covariance and update magnitude are adjusted according to the visual uncertainty. This division keeps learned temporal representation and explicit state estimation separate and inspectable.

Chapter 3 FieldAnchor Dataset and Benchmark Protocol

3.1 Data Collection and Route Split

FieldAnchor contains real UAV imagery acquired over agricultural land together with one registered satellite orthomosaic. The retained frames are organized into three flight routes, with partial spatial overlap among the routes. Route A is used for training, while Routes B and C are used for testing. This route-based protocol evaluates the localization performance of the model trained on Route A when applied to Routes B and C within the same mapped agricultural environment.

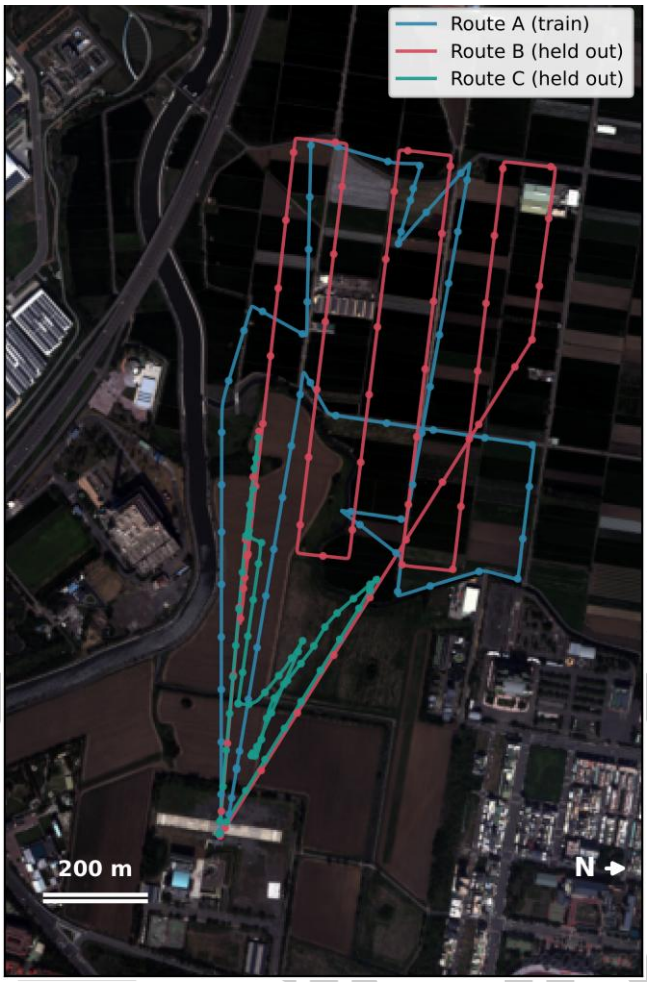

Figure 1. Route-held-out FieldAnchor split over the agricultural orthomosaic. The route-held-out split uses Route A for training, whereas Routes B and C are reserved for held-out evaluation. The separation is performed at the route level rather than by randomly mixing neighboring frames, so the held-out results measure localization on flight segments that are not part of the training trajectory. The same split is reused by the visual, temporal, and final-estimator experiments in Chapter 5.

Table **錯誤! 找不到參照來源。** defines the route-level split used throughout the thesis. The Split column separates model-development and held-out partitions, Route identifies the fli ght trajectory, Frames gives the number of retained images after filtering, and Role states how each subset is used. Because this split is fixed once at the route level, all later ablations and co

mparisons use the same training and held-out frames rather than regenerating a different partiti on for each experiment.

Table 1. FieldAnchor route statistics after flight-height filtering. Route A is the only training route. Routes B and C are reserved for held-out evaluation; their 3,534 frames form the main B+C test set used for aggregate reporting in Chapter 5.

Split	Route	Frames	Role
Train	A	1,732	Visual-model training
Test	B	2,276	Held-out evaluation
Test	C	1,258	Held-out evaluation
Total	A+B+C	5,266	Registered real-flight frames

As shown in Table **錯誤! 找不到參照來源。**, Route A contains 1,732 training frames, w hile Routes B and C contain 2,276 and 1,258 held-out frames, respectively. The two held-out r outes therefore provide 3,534 evaluation frames, and the complete registered dataset contains 5,266 retained frames. This fixed route-level allocation is reused throughout Chapter 5, so cha nges in experimental results cannot be attributed to a changing train/test split.

3.2 Fixed Satellite Anchor Lattice

The satellite orthomosaic is partitioned once into a permanent anchor lattice. Each anchor has a stable identifier, grid row and column, 320×320 pixel crop, metric center, and cached frozen backbone feature. The lattice stride is 32 pixels, providing a dense spatial response fiel d while allowing every UAV frame to query the same database. The released lattice contains 9 5,480 anchors. With the archived map registration, one 32-pixel gallery step corresponds to ap proximately one local lattice cell (about 4.75 m in the current experiment configuration), whic h provides the metric sampling scale used when constructing the forward local candidate field.

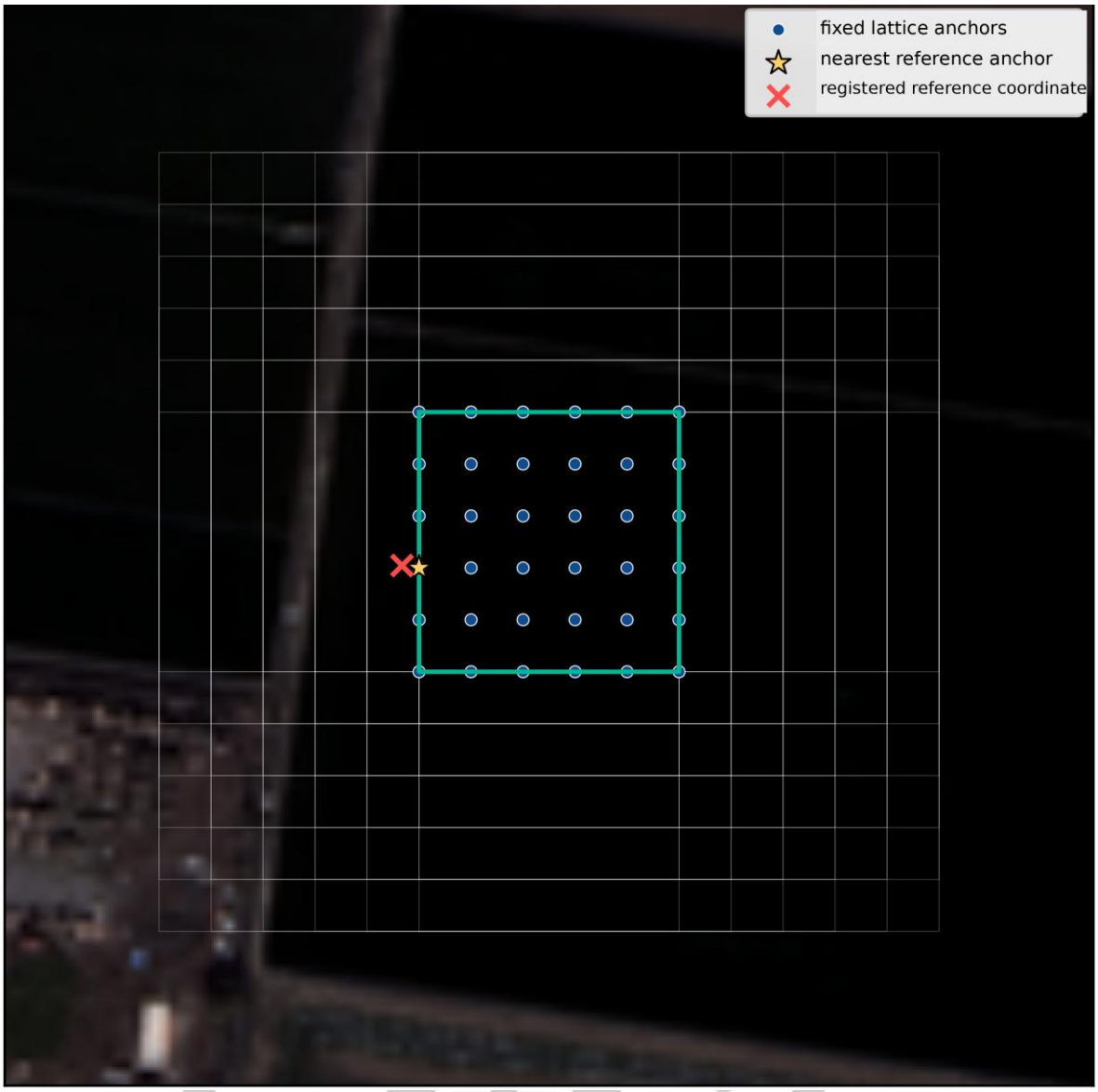

Figure 2. Permanent satellite lattice and a controlled local candidate window. The permanent satellite anchor lattice is fixed globally, while each query opens only one controlled local candidate window around the supplied coarse position estimate. The lattice provides stable anchor identities, metric centers, and cached visual features; the per-frame window only selects a compact subset from that permanent gallery. This separation keeps the candidate database identical across methods and evaluation runs.

3.3 Controlled Local Candidate Construction

A local-candidate manifest is generated before model training and evaluation so that ever y method receives an identical geometric search condition. Each retained UAV observation is associated with a predefined route reference point on the known planned trajectory. For the co ntrolled local-search benchmark, a coarse local position estimate is formed by applying the do cumented deterministic spatial perturbation to the associated reference point. This makes the l ocal search center intentionally imperfect while keeping the candidate geometry repeatable acr oss methods. The predefined route reference point is stored separately for supervision and eval

uation. Given the coarse estimate, the nearest permanent satellite anchor is found and one cont iguous 6×6 local lattice window is recorded with stable anchor identifiers, metric centers, and cached descriptors. Neither the coarse estimate nor the predefined route reference point is con catenated with the UAV or satellite visual embedding.

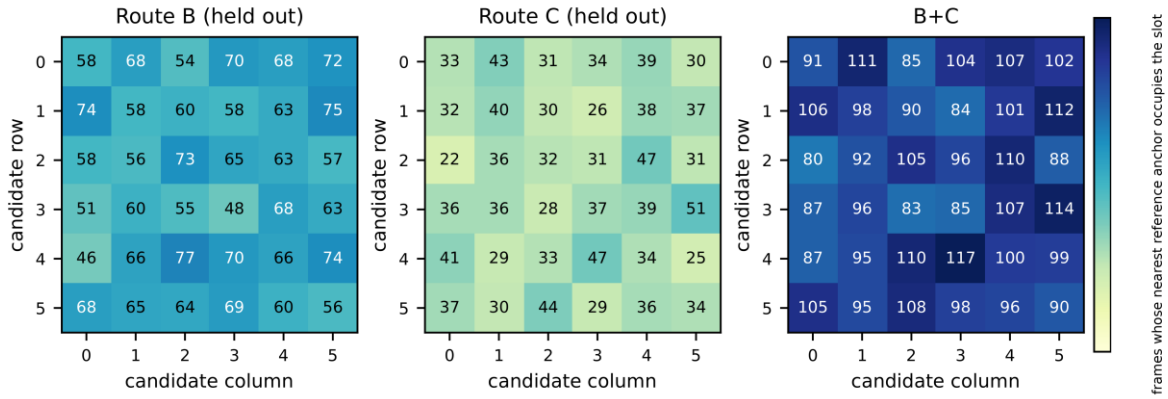

Figure 3. Reference-anchor slot distributions in the local candidate grid. The reference-anchor slot distribution spans the local grid rather than concentrating at one fixed array location. This verifies that the controlled local-search task cannot be solved by memorizing a constant candidate index: the localizer must still use visual evidence together with the candidate geometry to identify the correct region.

Algorithm 1 consolidates the controlled local candidate construction into two explicit sta ges. Step 1 maps the supplied coarse local position estimate to the nearest anchor in the perma nent satellite lattice. Step 2 then constructs the repeatable 6×6 neighborhood and retrieves the stable anchor identifier, metric center, and cached satellite descriptor for every candidate. The resulting object is the complete 36-candidate geometric neighborhood; the hard forward 3×6 r estriction is applied later in Section 4.3 before online satellite projection and similarity scoring. In this way, the coarse estimate determines only where the local gallery is opened and is not u sed as a visual feature or reported as the localization output.

For Algorithm 1, GridIndex denotes the deterministic conversion from the nearest perma nent anchor to its lattice row and column indices, and RegularGrid denotes construction of the fixed local candidate window centered on those indices. These are geometric indexing operati ons with no learned parameters.

Algorithm 1: Controlled Local Candidate Construction

Input: coarse local position estimate c_t^{coarse}, fixed satellite lattice G, and lattice stride;

Output: ordered local candidate set $C_t^{6\times6}$ with metric centers and cached descriptors;

// Step 1: Locate the coarse-position anchor in the permanent lattice

1 $i_0 \leftarrow \mathrm{argmin}_i \| c_i - c_t^{coarse} \|_2$;

2 $(row_0, col_0) \leftarrow$ GridIndex(i_0);

// Step 2: Construct one repeatable local candidate window

3 $C_t^{6\times6} \leftarrow$ RegularGrid($row_0, col_0, 6, 6, stride$);

4 **for each** $i \in C_t^{6\times6}$ **do**

5 read anchor ID i, metric center c_i, and cached satellite descriptor f_i^s;

6 **end for**

7 **return** $C_t^{6\times6}$ **and** c_i, f_i^s;

Step 1 of Algorithm 1 corresponds to Lines 1–2: the nearest permanent anchor is located in metric space and converted to its lattice row and column. Step 2 corresponds to Lines 3–6: t he regular local grid is instantiated around that anchor and the candidate metadata needed by t he later visual localizer are read without recomputing the frozen satellite backbone. Line 7 retu rns the complete local candidate object used by the forward-selection stage in Chapter 4.

Together, Figure 3 and Algorithm 1 describe complementary aspects of the same benchm ark operation. The figure verifies that the reference region can occur at different slots within th e local grid, while the algorithm specifies how that grid is reproduced deterministically for eve ry frame. This establishes a fixed geometric input condition before any learned similarity scori ng is performed.

3.4 Scope and Data Release

FieldAnchor provides the image, map-registration, route-split, anchor-lattice, planned-rou te reference-point, and local-candidate objects required for reproducible visual relocalization. Chapter 4 extends these dataset objects with the FieldAnchor-LR temporal model, inertial pred iction, uncertainty estimation, and Kalman filtering. This separation keeps the dataset protocol reusable while allowing the localization model to evolve independently.

The dataset package and the final FieldAnchor-LR model share the same route split, map registration, satellite lattice, planned-route geometry, controlled coarse-position condition, and predefined route reference points. The coarse-position condition defines only the candidate neighborhood, whereas the reference points are used for supervised targets and evaluation metrics. Model-specific temporal states are introduced only in Chapter 4 and are not stored as visual input annotations.

3.5 Acquisition Filtering and Image Preprocessing

The benchmark preparation process separates geometric information from visual model inputs. The retained frame list is first determined from the registered flight sequence and the documented quality filter. Each retained UAV image is center-cropped to 256×256 pixels without resizing the satellite crop. This gives a stable input field of view across training and testing, and prevents a change of image scale from being confounded with a change of candidate geometry.

Satellite patches are cropped directly from the registered orthomosaic. The 320×320 pixel patch size is intentionally larger than the 32-pixel lattice stride, so adjacent anchors have substantial image overlap. The overlap is not treated as independent additional imagery. It creates a locally continuous spatial response field in which a correct visual region should receive support from nearby anchors.

The same preprocessing is used in training and held-out evaluation: retained UAV frames are center-cropped to 256×256 pixels, satellite candidates are cropped directly from the orthomosaic at 320×320 pixels, and the fixed lattice uses a 32-pixel stride. Registration coordinates are never concatenated with the visual feature vector.

The archived frame list, crop origin, resize rule, and lattice metadata are released with the protocol so that evaluation does not silently change the visual evidence associated with an anchor.

3.6 Map Registration and Metric Coordinate System

FieldAnchor associates each retained UAV frame and satellite anchor with a common local metric coordinate system. Image-grid coordinates are useful for extracting satellite crops, but all localization errors and mean-shift kernels are computed in meters. This prevents the kernel bandwidth from depending on orthomosaic pixel resolution or image export settings.

Let $tilde(u)_i = [u_i, v_i, 1]^\top$ denote the homogeneous pixel coordinate of satellite anchor i, l et $c_i \in R^2$ denote its local metric center, and let $\prod$ denote the fixed pixel-to-metric registratio n matrix. The stored metric center is defined by

$$c_i = \Pi tilde(u)_i, \qquad tilde(u)_i = [u_i, v_i, 1]^\top. \tag{1}$$

The matrix $\prod$ is fixed for the archived orthomosaic derivative. The operational model do es not estimate $\prod$; it only uses the metric centers already stored in the anchor lattice. By isolat ing registration from visual similarity learning, the evaluation can distinguish map geometry e rrors from response-ranking errors.

All metric computations occur after registration. The model sees satellite pixels and cach ed visual descriptors, while the decoder reads only the candidate centers associated with those descriptors.

This representation also clarifies the scope of the reported error. Mean Localization Error (MLE) measures the Euclidean distance between the predicted metric position and the corresp onding predefined route reference coordinate. It is a visual-localization error measure rather th an a GPS receiver-accuracy measure.

3.7 Benchmark Package, Privacy Boundary, and Versioning

A reproducible local-localization benchmark requires more than image folders. The essen tial objects are the fixed lattice, route split files, candidate manifests, feature-cache schema, m odel configuration, and metric implementation. Together they ensure that a future model is eva luated on the same 36 local candidates for each frame rather than on a newly generated neighb orhood with different coverage.

The package layout is designed so that the response decoder can be reproduced without e xposing raw flight logs or personal source-machine paths.

Raw videos, sensor logs, timestamps, yaw values, altitude records, source-machine paths, and unredacted geographic metadata are outside the visual-anchor experiment. Keeping them separate limits avoidable privacy and location leakage; any public release should document pe rmissions, orthomosaic provenance, redaction, licensing, and dataset version hashes.

Chapter 4 Proposed Method

4.1 Overview

Let t denote the current UAV frame index. The system receives the current UAV image a nd the two preceding UAV images, a georeferenced satellite orthomosaic, the known planned r oute, and a controlled coarse local position estimate c_t^{coarse} for the current query. The coarse es timate defines only the local satellite search neighborhood. The visual encoder and recurrent r epresentation remain free of explicit map coordinates, while the final estimator operates in con tinuous route coordinates and returns a two-dimensional metric map position.

The FieldAnchor Local Relocalization (FieldAnchor-LR) framework contains six stages. A frozen MobileCLIP2-S2 encoder and two trainable view-specific projectors first construct c ross-view embeddings. A complete 6×6 local satellite geometry is formed. Before online sate llite projection and similarity scoring, a hard candidate restriction retains only the forward 3×6 subset; the rear half is excluded. Only the retained candidates are scored. Mean-Shift (MS) co nverts the visual similarities into a continuous visual observation and response uncertainty. A t hree-frame Gated Recurrent Unit (GRU) then estimates motion and measurement information from recent visual features and the previous state. The estimated velocity and acceleration for m a second-order inertial polynomial. Finally, an external uncertainty-aware Kalman Filter (K F) fuses the motion prediction with the MS visual measurement in continuous route coordinate s.

The term “Hard” refers to the explicit forward candidate restriction applied before Mean-Shift decoding: only the route-aligned forward 3×6 subset is retained from the complete 6×6 l ocal candidate geometry. It does not denote a separate variant of the Mean-Shift algorithm. Ac cordingly, “Hard” is used only for this candidate-support decision, whereas the spatial decoder itself is referred to uniformly as Mean-Shift (MS) throughout the thesis.

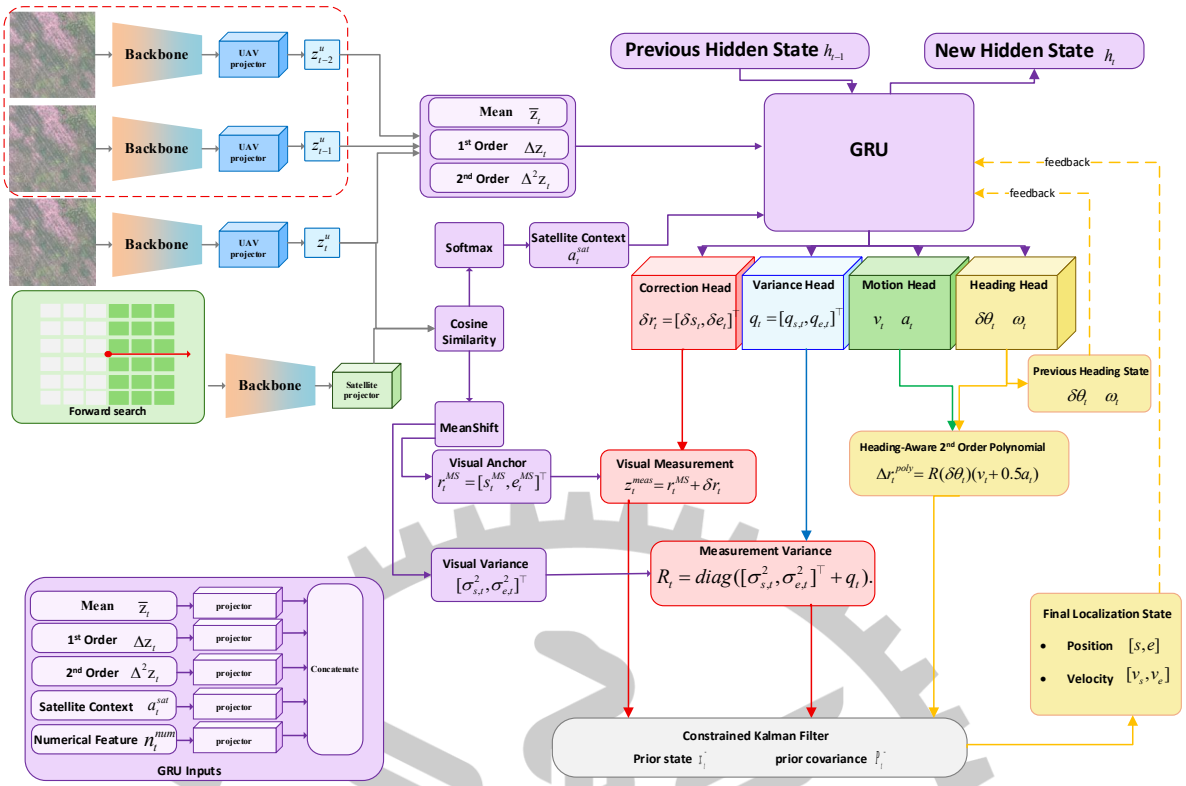

Figure 4. Overview of the FieldAnchor-LR local relocalization framework. The processing sequence is read from local-gallery construction to final state estimation. The coarse local position estimate terminates at candidate construction and is not appended to the UAV embedding, satellite embedding, or recurrent feature vector. The complete 6×6 geometry is first reduced by the hard forward 3×6 candidate restriction; visual matching is then performed only on the retained 18 candidates. Mean-Shift converts the resulting response field into a continuous observation, the GRU estimates temporal motion and measurement information, and the external Kalman Filter fuses the motion prediction with the visual measurement to produce the final continuous route state.

Each major block therefore has a single handoff to the next stage: visual representation pr oduces candidate response scores, spatial decoding produces a metric observation and uncertai nty, recurrent modeling produces motion and correction terms, and filtering produces the final route state. Presenting the method in this sequence keeps the learned visual representation, lea rned temporal state, and explicit estimator conceptually separate while still allowing them to o perate as one causal localization pipeline.

4.2 UAV–Satellite Visual Representation

Before defining the similarity score, let $E(\cdot)$ denote the frozen MobileCLIP2-S2 image e ncoder [12], let $P_u(\cdot)$ and $P_s(\cdot)$ denote the trainable UAV and satellite projection heads, respec tively, let I_t^u denote the current UAV image, and let $I_{t,i}^s$ denote satellite candidate i . The corr esponding L_2 -normalized 512-dimensional embeddings z_t^u and $z_{t,i}^s$ are defined as

$$z_t^u = \frac{P_u(E(I_t^u))}{\|P_u(E(I^t))\|_2}, \qquad z_{t,i}^s = \frac{P_s(E(I_{t,i}^s))}{\|P_s(E(I_{t,i}^s))\|_2}, \tag{2}$$

where the frozen encoder supplies a shared pretrained representation and the two independent projection heads adapt that representation to UAV-view and satellite-view appearance. Satellit e backbone descriptors are computed once and cached, whereas the lightweight satellite projec tor is evaluated only for the retained online candidates.

Let α denote the learnable logit-scale parameter. The cosine-similarity logit $l_{t,i}$ of cand idate i is

$$l_{t,i} = \exp(\alpha)(z_t^u)^\top z_{t,i}^s. \tag{3}$$

Before using the candidate posterior, let $\tau > 0$ denote the response temperature and let N denote the number of retained candidates. The normalized visual posterior $p_{t,i}$ is

$$p_{t,i} = \frac{\exp(\frac{l_{t,i}}{\tau})}{\sum_{j=1}^{N}(\exp)(\frac{l_{t,j}}{\tau})}, \qquad \sum_{j=1}^{N}(p_{t,i}) = 1, \tag{4}$$

where the posterior sums to one over the retained local candidates. FieldAnchor-LR uses $\tau = 0.30$. The same posterior supplies the response mass for MS and the satellite context used b y the temporal model. In Algorithm 2, this normalization is the first operation of Step 1, after which each retained candidate center is initialized as an independent Mean-Shift seed.

4.3 Hard Forward 3×6 Candidate Restriction

The candidate geometry is first constructed as a complete 6×6 neighborhood containing 36 candidates around the supplied coarse local position estimate. Let $c_t \in R^2$ denote the neigh borhood center, let $c_{t,i} \in R^2$ denote candidate center i, and let θ_t denote the causal search d irection. Define the forward unit vector $f_t = [\cos\theta_t, \sin\theta_t]^\top$ and the perpendicular unit vector $n_t = [-\sin\theta_t, \cos\theta_t]^\top$. The longitudinal and lateral projections are

$$d_{t,i}^{\parallel} = (c_{t,i} - c_t)^\top f_t, \qquad d_{t,i}^{\perp} = (c_{t,i} - c_t)^\top n_t, \tag{5}$$

where $d_{t,i}^{\parallel}$ measures the forward displacement and $d_{t,i}^{\perp}$ measures the lateral displacement. The 18 candidates with the largest forward projection are retained and ordered from front to back a nd left to right. This discrete support selection is the “Hard” operation in FieldAnchor-LR: the remaining 18 rear candidates are removed before online satellite projection, cosine-similarity scoring, and Mean-Shift decoding. Only the retained 18 satellite descriptors are projected and scored. The direction is causal because it combines the local route tangent with the previously estimated heading residual and does not use any future frame.

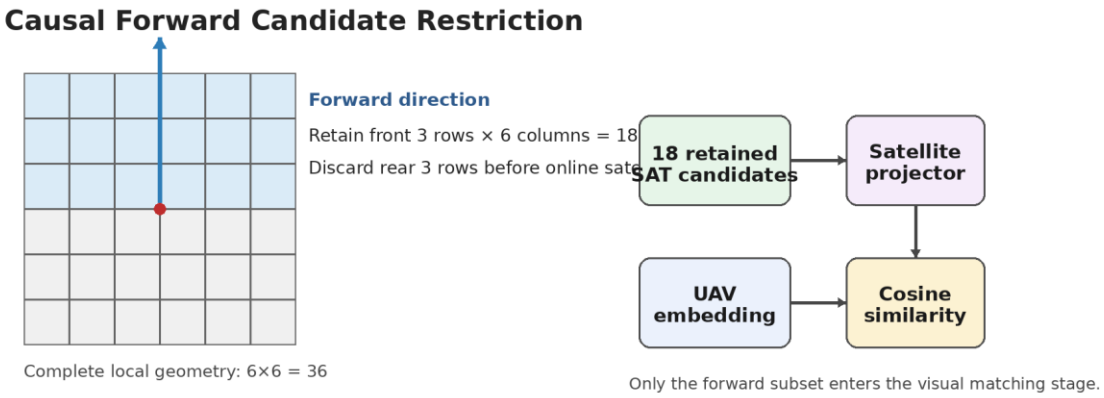

Figure 5. Hard forward 3×6 candidate restriction. The diagram gives the geometric interpretation of Equation (5): the complete 6×6 local neighborhood is projected into route-aligned longitudinal and lateral coordinates, after which the hard restriction retains only the causal forward 3×6 half-grid. These retained 18 candidates form the response field used by the subsequent visual decoder, while candidates behind the current motion direction are excluded from online scoring.

The hard forward restriction is causal: the orientation of the retained half-grid is determin ed from the planned-route tangent and the previously estimated heading residual, not from a fu ture UAV frame. Consequently, the candidate reduction changes the spatial support and online scoring cost without introducing future-frame information into the localizer.

4.4 Mean-Shift Visual Localization

After the hard forward restriction, $N=18$ metric candidate centers remain. Mean-Shift (MS) [11] is used as a continuous spatial decoder rather than a winner-take-all anchor selector. Let $m_{t,i}^{(k)}\in\mathbb{R}^2$ denote the mode initialized from candidate i after k updates, let $h>0$ denote t he Gaussian kernel bandwidth, and let K denote the number of iterations. The initialization is $m_{t,i}^{(U)}\in c_{t,i}$. Each mode is updated by

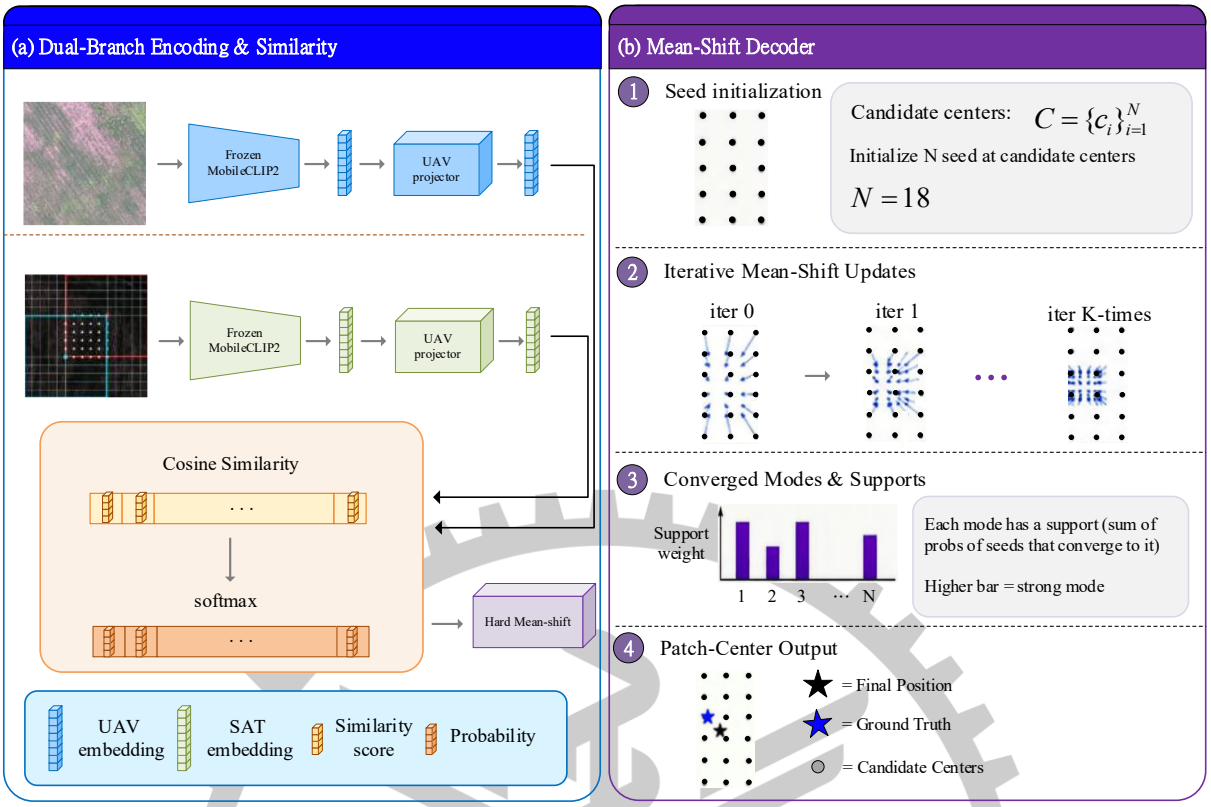

Figure 6. Mean-Shift (MS) continuous visual localization flow. The three visualized operations correspond to Algorithm 2. Candidate logits are normalized and candidate centers initialize the seeds; the seeds are iteratively shifted toward locally supported response mass; and the converged modes are weighted to form the continuous visual coordinate and uncertainty. The figure therefore visualizes the same computation defined algebraically in Equations (6)–(9).

$$m_{t,i}^{(k+1)} = \frac{\sum_{j=1}^{N} p_{t,j} \exp\left(-\frac{\left\|m_{t,i}^{(k)} - c_{t,j}\right\|_2^2}{2h^2}\right) c_{t,j}}{\sum_{j=1}^{N} p_{t,j} \exp\left(-\frac{\left\|m_{t,i}^{(k)} - c_{t,j}\right\|_2^2}{2h^2}\right)}, \tag{6}$$

where $h = 8$ m and $K = 3$ in FieldAnchor-LR. Every candidate seed remains active during the update, allowing nearby candidates with consistent visual support to converge toward the sam e spatial mode. This iterative update is Step 2 of Algorithm 2 (Lines 3–8); the nested loops ma ke explicit that every seed is repeatedly updated using the response mass from all retained can didates inside the Gaussian kernel.

The denominator normalizes the kernel-weighted posterior mass, whereas the numerator is the corresponding weighted sum of candidate centers. These terms do not cancel because ea ch weight in the numerator is multiplied by a different candidate center before summation. Eq uation (6) therefore returns a weighted average location for each Mean-Shift seed rather than r educing to a single candidate center.

Let $\rho_{t,i}$ denote the density support of the i-th converged mode. It is defined by

$$\rho_{t,i} = \sum_{j=1}^{N} p_{t,j} \exp\left(-\frac{\left\| m_{t,j}^{(k)} - c_{t,j} \right\|_2^2}{2h^2} \right). \tag{7}$$

Let $\beta > 0$ denote the mode-density temperature and let $\omega_{t,i}$ denote the normalized mode weight. FieldAnchor-LR uses $\beta = 12$, and the weights are

$$\omega_{t,i} = \frac{\exp(\beta \rho_{t,i})}{\sum_{r=1}^{N} \exp(\beta \rho_{t,r})}, \qquad \sum_{i=1}^{N} \omega_{t,i} = 1. \tag{8}$$

The final continuous MS visual coordinate x_t^{MS} is

$$x_t^{MS} = \sum_{i=1}^{N} \omega_{t,i} m_{t,i}^{(K)}, \tag{9}$$

so the visual observation is not snapped to a discrete satellite anchor. Equations (7)–(9) constitute Step 3 of Algorithm 2 (Lines 9–13): density support is computed for each converged mode, the supports are converted to normalized mode weights, and the final continuous position is formed as their weighted combination. The same converged mode distribution is retained to estimate visual measurement uncertainty before temporal fusion.

Algorithm 2 consolidates the response normalization in Equation (4) and the continuous Mean-Shift operations in Equations (6)–(9) into one executable sequence. Unlike a discrete Top-1 selector, the procedure keeps all candidate seeds active during mode seeking and returns both the continuous visual coordinate and the mode-space uncertainty required by the later state estimator.

For Algorithm 2, Softmax denotes the normalized exponential operation already defined by Equations (4) and (8). WeightedModeVariance denotes the weighted second-moment calculation that uses the converged modes and normalized mode weights to obtain the mode-space variance described in Section 4.8. These names are shorthand for the stated calculations rather than additional learned modules.

Algorithm 2: Mean-Shift (MS) Visual Localization

Input: candidate logits l_i, metric centers c_i, temperature τ, bandwidth h, iterations K, and densit y scale β;

Output: continuous visual coordinate x^{MS}, mode-space variance $\sum^{MS}$, and mode weights ω_i;

// Step 1: Normalize the visual response and initialize the seeds

1 $p \leftarrow \text{Softmax}(\frac{l}{\tau})$;

2 **for** $i \leftarrow 1$ **to** N **do** $m_i \leftarrow c_i$; **end for**

// Step 2: Iteratively shift every seed toward local response mass

3 **for** $k \leftarrow 0$ **to K-1 do**

4 **for** $i \leftarrow 1$ **to** N **do**

5 $w_{ij} \leftarrow p_j \exp(-\frac{\| m_i - c_j \|_2^2}{2h^2})$ for all j;

6 $m_i \leftarrow (\frac{\sum_j (w_{ij}) c_j}{\sum_{j(w_{ij})}})$;

7 **end for**

8 **end for**

// Step 3: Density-weight the converged modes and form the output

9 $\rho_i \leftarrow \sum_j (p_j) \exp(-\frac{\| m_i - c_j \|_2^2}{2h^2})$;

10 $\omega \leftarrow \text{Softmax}(\beta\rho)$;

11 $x^{MS} \leftarrow \sum_i (\omega_i) m_i$;

12 $\sum^{MS} \leftarrow \text{WeightedModeVariance}(m_i, \omega_i)$;

13 **return** x^{MS}, $\sum^{MS}$, **and** ω_i;

Reading Algorithm 2 sequentially, Lines 1–2 establish the normalized response field and seed locations, Lines 3–8 perform the iterative local mode search, and Lines 9–13 convert the converged modes into a single visual coordinate and uncertainty estimate. The output x_t^{MS} is p rojected into the continuous route coordinate system in Section 4.5, whereas $\sum_t^{MS}$ is propagate d to the uncertainty model in Section 4.8. Thus the algorithm is the direct bridge between fram e-level visual similarity and the temporal estimator.

4.5 Continuous Route Coordinates and Waypoint Representation

The temporal estimator uses continuous route coordinates so that forward progress and la teral displacement retain clear physical meaning. Let q_k and q_{k+1} denote consecutive predefin ed route reference points, let $L_k = \| q_{k+1} - q_k \|_2$ denote the segment length, let $u_k = \frac{q_{k+1} - q_k}{L_k}$ de note the segment unit vector, let $n_k = [-u_k, y, u_k, x]^\top$ denote the cross-track unit vector, and let

S_k denote the accumulated route length at q_k . For route state $r=[s,e]^\top$ on segment k , the metric coordinate is

$$x(s,e)=q_k+(s-S_k)u_k+en_k, \tag{10}$$

where *s* is along-route progress and *e* is signed cross-track displacement. Around a waypoint, an instantaneous switch of the cross-track axis would rotate the same nonzero lateral state disc ontinuously. FieldAnchor-LR therefore rotates the local route frame causally after the filtered progress passes the waypoint, which avoids premature turning while preserving a continuous XY trajectory. The MS position is projected to this route coordinate before recurrent state esti mation.

4.6 Three-Frame Gated Recurrent Unit Temporal Modeling

Let z_t , z_{t-1}, and z_{t-2} denote the current and two preceding UAV embeddings after the vi ew-specific projection head. The three-frame representation uses a temporal mean, the most re cent first difference, and the corresponding second difference, defined by

$$\bar{z}_t=\frac{z_{t-2}+z_{t-1}+z_t}{3}, \qquad \Delta z_t=z_t-z_{t-1}, \qquad \Delta^2 z_t=(z_t-z_{t-1})\text{-}(z_{t-1}-z_{t-2}), \tag{11}$$

The first difference captures the most recent change in visual representation, whereas the second difference describes how that change itself evolves. In the notation used below, the sup erscript labels “sat”, “vis”, and “num” are descriptive abbreviations for satellite, visual, and nu meric, respectively; they identify the role of a quantity and do not denote exponentiation. The satellite context is formed by posterior-weighting the retained satellite embeddings with the no rmalized candidate posterior from Equation (4). The visual innovation is the route-coordinate difference between the current projected Mean-Shift observation and the one-step causal moti on prior available before Kalman fusion. The two mode-space variance components used by th e numeric state are produced by the Mean-Shift decoder and are written explicitly in Equation (17). The scalar turn-rate component of the heading state is distinct from the indexed Mean-Sh ift mode weights defined in Equation (8).

Let a_t^{sat} denote the posterior-weighted satellite context, let δ_t^{vis} denote the innovation bet ween the current MS observation and the motion prediction, let v_{t-1} denote the previous route velocity, and let $h_{t-1}^{\theta}=[\delta\theta_{t-1},\omega_{t-1}]^\top$ denote the previous heading residual and turn-rate state. Th e eight-dimensional numeric state n_t^{num} contains the two log mode-space variances, the two-di

mensional visual innovation, the two-dimensional previous velocity, and the two-dimensional heading state h_{t-1}^{θ} .

The five projection symbols introduced below denote learned feature-projection blocks a pplied separately to the temporal mean, first difference, second difference, satellite context, an d numeric state. Their role is to place these heterogeneous inputs in compatible feature spaces before fusion; they do not create additional route states or geometric measurements. For Equat ion (12), the semicolon-separated bracket notation is defined as feature concatenation, meanin g that the projected feature blocks are appended along the feature dimension before being pass ed to the recurrent unit.

Let Φ_{μ} , Φ_{Δ}, Φ_{Δ}^{2} , Φ_{s}, and Φ_{n} denote learnable projection functions for the temporal m ean, first difference, second difference, satellite context, and numeric state. The recurrent inpu t g_t is

$$g_t = [\Phi_{\mu}(\bar{z}_t); \Phi_{\Delta}(\Delta z_t); \Phi_{\Delta}^{2}(\Delta^{2} z_t); \Phi_{s}(a^{t}); \Phi_{n}(n_t^{num})], \tag{12}$$

The operator $[\cdot;\cdot]$ denotes feature concatenation. Let h_{t-1} denote the previous recurrent hi dden vector. The current hidden state is

$$h_t = \mathrm{GRUCell}(g_t, h_{t-1}), \tag{13}$$

The GRU hidden dimension is 256. Four lightweight output heads decode the hidden stat e into route velocity and acceleration, heading residual and turn rate, visual-measurement corr ection, and positive extra measurement variance. The heading variables remain internal motio n states and are not treated as a separate final localization output.

The GRUCell in Equation (13) follows the standard Gated Recurrent Unit formulation [1 3]. It computes reset and update gates with sigmoid activation from the current recurrent input and previous hidden vector, forms a candidate hidden representation using a hyperbolic-tange nt activation with reset-gated history, and then interpolates between the previous and candidat e hidden representations. This recurrent update is causal and uses no future-frame information.

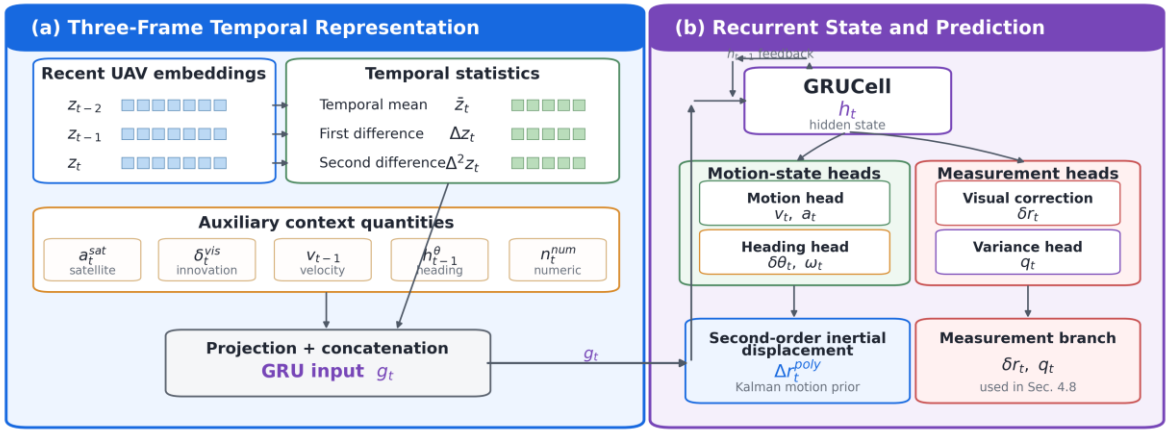

Figure 7. Three-frame GRU and second-order inertial prediction.

The architecture in Figure 7 groups the temporal estimator into two coupled parts. The three recent UAV em beddings first form the temporal mean, first difference, and second difference; these features are combined with the satellite context, visual innovation, previous route velocity, previous heading state, and the numeric state to construct the GRU input. The recurrent hidden state is then decoded by separate motion, heading, v isual-correction, and variance heads. Only the motion and heading outputs form the second-order inertial di splacement, whereas the visual-correction and variance outputs are passed to the measurement branch descr ibed in Section 4.8.

This arrangement also explains why three frames are used even though the final output is one position per frame. The current observation supplies the present visual evidence, while th e two preceding embeddings make both a recent change and a change-of-change observable to the GRU. The temporal ablation in Table **錯誤! 找不到參照來源。** later evaluates whether th is additional history improves held-out localization.

4.7 Second-Order Inertial Motion Prediction

The recurrent motion head produces route velocity $v_t = [v_t^{\parallel}, v_t^{\perp}]^{\top}$ and route acceleration $a_t = [a_t^{\parallel}, a_t^{\perp}]^{\top}$. Let $\delta\theta_t$ denote the causal heading residual, and let $R(\delta\theta_t)$ denote the corresp onding two-dimensional rotation matrix

$$R(\delta\theta_t) = [\cos\delta\theta_t, -\sin\delta\theta_t; \sin\delta\theta_t, \cos\delta\theta_t], \tag{14}$$

where $\delta\theta_t$ is estimated causally from the recurrent state together with the preceding temporal state. With one frame treated as one discrete time step, the second-order inertial displacement i s

$$\Delta r_t^{poly} = R(\delta\theta_t)(v_t + \frac{1}{2}a_t), \tag{15}$$

With one frame treated as one discrete time step, Equation (15) combines the GRU-predicted velocity and acceleration through the second-order polynomial and rotates the resulting displacement by the predicted heading residual. The resulting displacement is passed directly to the Kalman prediction as the learned motion prior for the current frame.

4.8 Visual Measurement Correction and Uncertainty Estimation

Let $r_t^{MS} = [s_t^{MS}, e_t^{MS}]^\top$ denote the route-coordinate projection of the continuous MS observation, and let $\delta r_t = [\delta s_t, \delta e_t]^\top$ denote the correction predicted by the GRU. The visual measurement supplied to the KF is

$$z_t^{meas} = r_t^{MS} + \delta r_t, \tag{16}$$

To quantify the visual response itself, let u_t and n_t denote the local route-parallel and cross-track unit vectors, let $m_{t,i}^{(K)}$ denote the converged MS modes, and let $\omega_{t,i}$ denote the mode weights. The mode-space variances are

$$\sigma_{s,t}^2 = \sum_i \omega_{t,i} [(m_{t,i}^{(K)} - x_t^{MS})^\top u_t]^2, \qquad \sigma_{e,t}^2 = \sum_i \omega_{t,i} [(m_{t,i}^{(K)} - x_t^{MS})^\top n_t]^2, \tag{17}$$

where tightly converged modes yield a small variance and spatially separated plausible modes increase the uncertainty. Let $q_t = [q_{s,t}, q_{e,t}]^\top$ denote the non-negative extra variance predicted by the GRU through a softplus output. The measurement covariance is

$$R_t = diag([\sigma_{s,t}^2, \sigma_{e,t}^2]^\top + q_t). \tag{18}$$

In Equation (18), softplus denotes the standard smooth positive mapping used to convert the variance-head output into the non-negative extra variance vector q_t. The function $diag(\cdot)$ is the diagonal-matrix operator, which places the elements of its vector argument on the main diagonal. Thus, the measurement covariance is formed directly from the Mean-Shift mode-space variance and the learned extra variance.

4.9 Uncertainty-Aware Kalman Filtering

The final estimator operates in route coordinates. Let $x_t = [s_t, e_t, v_{s,t}, v_{e,t}]^\top$ denote the KF state, let P_t denote its covariance, let F denote the constant-velocity transition matrix, let Q denote the process-noise covariance, let H denote the 2×4 position-observation matrix, and let R_t denote the measurement covariance defined above. The recurrent polynomial displacement is injected into the position prediction according to Equation (19). The four state componen

ts are ordered as along-route progress, cross-track displacement, along-route velocity, and cros s-track velocity.

$$x_t^- = Fx_{t-1} + b(\Delta r_t^{poly}), \qquad P_t^- = FP_{t-1}F^\top + Q, \tag{19}$$

The injection function $b(\cdot)$ inserts the two-dimensional polynomial displacement into the position components of the four-dimensional state. Let ν_t denote the visual innovation, let S_t denote the innovation covariance, and let η_t denote the Normalized Innovation Squared (NIS) statistic. They are

$$\nu_t = z_t^{meas} - Hx_t^-, \qquad S_t = HP_t^- H^\top + R_t, \qquad \eta_t = \nu_t^\top S_t^{-1} \nu_t, \tag{20}$$

where a large NIS indicates disagreement between the inertial prediction and the current visua l measurement. The NIS is retained as a consistency statistic describing agreement between th e inertial prediction and the current visual measurement, while the visual reliability entering th e Kalman gain is represented by the measurement covariance defined in Equation (18). Let K_t denote the Kalman gain and let I denote the 4×4 identity matrix. The covariance-stable upd ate is

$$K_t = P_t^- H^\top (HP_t^- H^\top + R_t)^{-1}, \qquad x_t = x_t^- + K_t \nu_t, \qquad P_t = (I - K_t H) P_t^- (I - K_t H)^\top + K_t R_t K_t^\top. \tag{21}$$

Equation (21) gives the posterior state and covariance used by the next frame. Route prog ress and cross-track displacement remain components of the same continuous route-coordinate state, while their updates are determined by the motion prediction, the visual measurement, an d the learned measurement covariance through the Kalman gain.

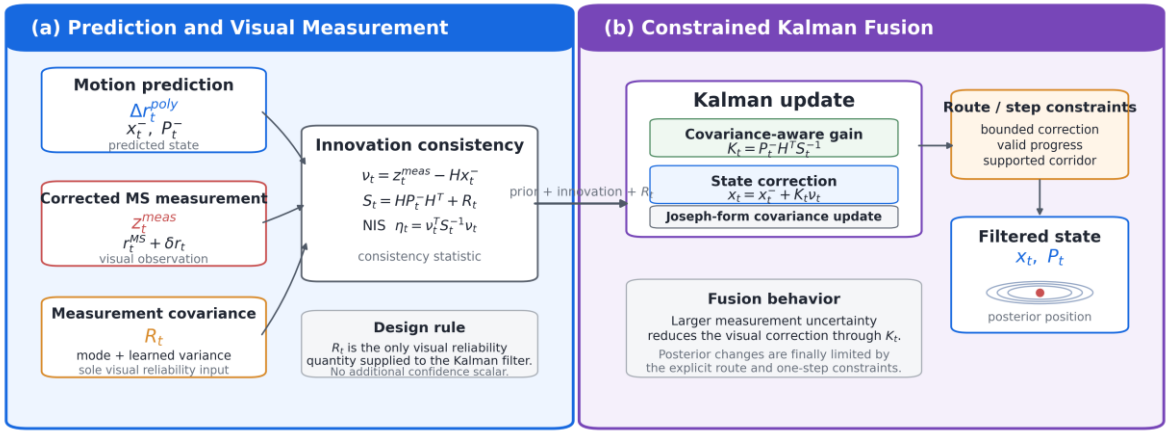

Figure 8. Uncertainty-aware Kalman fusion. Figure 8 separates the Kalman stage into prediction and measurement formation followed by uncertainty-aware fusion. The motion model supplies the prior state and covariance, the corrected Mean-Shift observation supplies the position measurement, and the measurement covariance R_t supplies the visual uncertainty used by the Kalman gain. Their disagreement is summarized by the innovation and NIS, after which the covariance-aware update produces the posterior state and covariance.

The diagram emphasizes that R_t is the measurement covariance and the sole visual reliability quantity supplied to the Kalman Filter. It enters the covariance-aware update directly, while the recurrent motion state remains a separate source of temporal prediction; no additional reliability input is supplied to the filter.

4.10 Training Objective

Training is divided into a visual stage and a temporal stage. In the visual stage, only Route A is used to optimize the two projection heads while the MobileCLIP2-S2 backbone remains frozen. Each training observation first retrieves the satellite candidates associated with its controlled coarse local position estimate. The candidate nearest the corresponding predefined route reference point provides the classification target, while the continuous MS coordinate is supervised directly against that reference point. Let L_{CE} denote cross-entropy over the local satellite candidates, let L_{coord} denote the Smooth L1 coordinate loss, and let L_{visual} denote the visual-stage objective. The objective is

$$L_{visual} = L_{CE} + 0.25 L_{coord}, \tag{22}$$

where label smoothing is applied to L_{CE}. The temporal stage is trained sequentially with Truncated Backpropagation Through Time (TBPTT) over contiguous Route-A frames under the same controlled local-search condition. Consecutive predefined route reference points provide the supervised motion targets, while the GRU itself receives recent visual features, satellite context, and previous model state. Let L_{meas} denote the Smooth L1 visual-measurement loss, L_{step} the Smooth L1 next-step displacement loss, L_{vel} a weak velocity loss, L_{head} the periodic heading loss, and L_{var} the Gaussian negative log-likelihood for measurement variance. The active temporal objective is

$$L_{temporal} = 1.00L_{meas} + 3.00L_{step} + 0.25L_{vel} + 1.25L_{head} + 0.05L_{var}. \tag{23}$$

Acceleration is not assigned an independent supervised term; it is learned through $L_{s}tep$ because it directly contributes to the second-order displacement. Turn rate, speed, route progress, and a separate cross-motion regularizer are not active independent losses in FieldAnchor-LR. At each TBPTT boundary the computational graph is detached, while the recurrent and filter states themselves remain continuous. The predefined route reference points used to form the losses are training annotations and are not concatenated with the visual embeddings or recurrent input vector. Algorithm 3 organizes these objectives into the same two stages described in the text: Lines 1–9 implement visual representation learning with Equation (22), and Lines 10–19 implement sequential temporal learning with Equation (23).

In Algorithm 3, LocalCandidates refers to Algorithm 1; ViewProjectors and ScaledCosine refer to the operations in Equations (2)–(4); MeanShift refers to Algorithm 2; ForwardCandidates refers to the hard forward 3×6 candidate restriction defined by Equation (5); VisualObservation denotes the continuous Mean-Shift measurement described in Sections 4.4 and 4.8; GRU denotes the recurrent computation in Equations (12)–(13); and KalmanFilter denotes the prediction and update defined in Equations (19)–(21). The function-style names are therefore compact references to operations already defined in the text, not unspecified black-box modules.

Algorithm 3: FieldAnchor-LR Two-Stage Training Procedure

Input: Route-A UAV frames, predefined Route-A reference points X_{ref}, coarse-position candidate manifests M, fixed satellite gallery G, and frozen MobileCLIP2-S2;
Output: trained UAV/satellite projectors and three-frame GRU parameters;

// Stage 1: visual representation learning
1 Freeze MobileCLIP2-S2 and cache satellite backbone descriptors;
2 **for each Route-A minibatch do**
3 $C \leftarrow$ LocalCandidates(coarse position estimate from M, fixed satellite lattice);
4 $(z^u, z^s) \leftarrow$ ViewProjectors(C,UAV image);
5 $l \leftarrow$ ScaledCosine(z^u, z^s);
6 $(x^{MS}, \sum^{MS}) \leftarrow$ MeanShift(l,candidate centers);
7 $L_{visual} \leftarrow L_{CE}$(nearest reference anchor) + 0.25 L_{coord}(x^{MS}, x_{ref});
8 Update UAV and satellite projection heads with AdamW;
9 **end for**
// Stage 2: sequential temporal learning
10 Freeze the visual localizer; initialize GRU hidden state and route-coordinate KF state;
11 **for** $t \leftarrow first$ **to last contiguous Route-A frame do**
12 $C_t \leftarrow$ ForwardCandidates(coarse position estimate from M,previous causal direction);
13 $(x_t^{MS}, \sum^{MS}, a_t^{sat}) \leftarrow$ VisualObservation(C_t);
14 $(v_t, a_t, \delta\theta_t, \delta r_t, q_t, h_t) \leftarrow$ GRU(three-frame features,previous state);
15 $\Delta r_t^{poly} \leftarrow R(\delta\theta_t)(v_t + 0.5a_t)$;
16 $x_t \leftarrow$ KalmanFilter($\Delta r_t^{poly}, r_t^{MS} + \delta r_t, R_t$);
17 accumulate $L_temporal$;
18 Every 32 frames: back-propagate, update GRU, and detach graph history;
19 **end for**
20 **return trained projectors and GRU parameters;**

Within the first stage of Algorithm 3, the frozen MobileCLIP2-S2 backbone and cached s atellite features are reused while only the view-specific projectors are optimized. Line 6 invok es the Mean-Shift visual localizer summarized in Algorithm 2, and Line 7 applies the visual ob jective in Equation (22). The second stage then freezes this visual localizer and trains the three -frame GRU sequentially, preserving the recurrent and filter states across contiguous frames w hile detaching only the graph history at the TBPTT boundary.

The two-stage structure prevents the temporal model from changing the underlying satell ite-response representation while it learns motion and measurement behavior. It also makes th e training procedure consistent with the inference pipeline: the visual stage produces the same MS observation used online, and the temporal stage learns how that observation interacts with the recurrent motion state and the external uncertainty-aware Kalman Filter.

4.11 End-to-End Inference

During held-out evaluation, UAV frames are processed sequentially with fixed model par ameters. For each query, the controlled benchmark supplies a coarse local position estimate th at establishes one complete 6×6 satellite neighborhood. The hard forward candidate restriction then retains only the route-aligned forward 3×6 subset, and the current UAV image is matched only against those 18 candidates. The previous two UAV embeddings, recurrent hidden state, motion state, and KF posterior are propagated from the preceding frame and are used for temp oral prediction and fusion; they do not replace the benchmark-defined candidate neighborhood. Predefined route reference points are read only after prediction to compute localization metric s. The full held-out localization sequence is summarized in Algorithm 4, which connects hard forward local visual matching, Mean-Shift measurement formation, GRU-based motion estim ation, inertial prediction, and uncertainty-aware Kalman fusion.

In Algorithm 4, RouteTangent denotes the planned-route tangent adjusted by the previous causal heading residual; Forward denotes the hard forward candidate restriction in Equation (5); UAVProjector and SatelliteProjector denote the view-specific projections in Equation (2); ScaledCosine denotes Equations (3)–(4); MeanShift denotes Algorithm 2; ProjectToRoute den otes the coordinate projection in Section 4.5; GRU denotes Equations (12)–(13); Measurement Covariance denotes Equations (17)–(18); KFPredict denotes Equation (19); KFUpdate denotes Equations (20)–(21); and RouteToMetricXY denotes conversion from route coordinates to me tric map coordinates using Equation (10).

Algorithm 4: FieldAnchor-LR Controlled Local Temporal Localization

Input: current UAV image I_t^u, two recent UAV embeddings, coarse local position estimate c_t^{coarse}, known planned route Q, cached satellite gallery, and previous GRU/KF states;

Output: final metric position p_t and updated recurrent/filter states;

// Hard forward local visual localization

1 $c_t \leftarrow c_t^{coarse}$;

2 $\theta_t \leftarrow$ RouteTangent(previous filtered progress)+previous heading residual;

3 $C_t \leftarrow$ Forward 3×6 ($c_t, \theta_{t,fixed}$,satellite lattice);

4 $z_t^u \leftarrow$ UAVProjector $(E(I_t^u))$;

5 $z_t^s \leftarrow$ SatelliteProjector(cached descriptors of C_t);

6 $l_t \leftarrow$ ScaledCosine(z_t^u, z_t^s);

// Continuous visual measurement and recurrent motion

7 $(x_t^{MS}, \sum_t^{MS}) \leftarrow$ MeanShift(l_t,centers(C_t));

8 $r_t^{MS} \leftarrow$ ProjectToRoute(x_t^{MS});

9 $a_t^{sat} \leftarrow \sum_i (p_{t,i}) z_{t,i}^s$;

10 $(v_t, a_t, \delta\theta_t, \delta r_t, q_t, h_t) \leftarrow$ GRU($z_t, z_{t-1}, z_{t-2}, a_s^{sat}$,previous state);

11 $\Delta r_t^{poly} \leftarrow R(\delta\theta_t)(v_t + 0.5a_t)$;

12 $z_t^{meas} \leftarrow r_t^{MS} + \delta r_t$;

// Uncertainty-aware state fusion

13 $R_t \leftarrow$ MeasurementCovariance($\sum_t^{MS}, q_t$);

14 $x_t^- \leftarrow$ KFPredict(previous state, Δr_t^{poly});

15 $x_t \leftarrow$ KFUpdate(x_t^-, z_t^{meas}, R_t);

16 $p_t \leftarrow$ RouteToMetricXY(x_t);

17 Update recent embeddings, recurrent state, and KF state;

18 **return p_t and updated states;**

Algorithm 4 is the operational counterpart of Sections 4.2–4.9. Lines 1–6 correspond to l ocal candidate construction, hard forward restriction, visual projection, and similarity scoring; Lines 7–12 convert the response into an MS observation, project it to route coordinates, and o btain the GRU motion and measurement terms; and Lines 13–18 form the measurement covari ance, execute the Kalman predict/update cycle, convert the posterior route state to metric coor dinates, update the causal temporal state, and return the final metric position.

This step-by-step correspondence is intentional: Figure 4 gives the global block diagram, Figures 5–8 explain the geometry and state flow inside the major blocks, Equations (2)–(21) d efine the numerical operations, and Algorithm 4 records their causal execution order for one h eld-out frame. These elements therefore describe different levels of the same pipeline rather th an independent procedures.

4.12 Computational Complexity and Reproducibility

For N retained candidates and K Mean-Shift iterations, each active seed evaluates a kerne l against all retained candidates at every iteration, giving a time complexity of $O(KN^2)$. In Fiel dAnchor-LR, N = 18 and K = 3, so the Mean-Shift decoder performs $3 \times 18^2 = 972$ pairwise k ernel evaluations per frame.

Using the complete 6 × 6 neighborhood under the same three iterations would require $3 \times 36^2 = 3{,}888$ pairwise evaluations. Equivalently, the forward 3 × 6 restriction reduces the Mean -Shift kernel workload by 75% and also halves the number of online satellite projections and c osine-similarity evaluations from 36 candidates to 18.

The repository stores the visual and temporal checkpoints separately, caches the frozen sa tellite backbone descriptors, and records route-wise prediction files and experiment summarie s. Quantitative values are reported only when they are present in versioned outputs; unexecute d settings are marked with an em dash rather than being estimated.

Chapter 5 Experiments and Results

5.1 Experimental Settings and Metrics

All experiments retain the FieldAnchor route split: Route A is used for visual and tempor al model training, while Routes B and C are held out for final evaluation. For every query, the same controlled coarse-position condition and fixed satellite lattice are used to establish the co mplete 6×6 local candidate neighborhood. The predefined route reference point remains separ ate and is used only to form supervised targets on Route A and to compute metrics on the held -out routes. All methods in a given internal comparison use the same route split, satellite galler y, candidate construction rule, and metric coordinate system. Accordingly, the reported FieldA nchor-LR results quantify local visual refinement around predefined planned-route reference p oints rather than unrestricted map-wide acquisition.

The primary localization metrics are Mean Localization Error (MLE), Median Localizati on Error (MedLE), the 90th and 95th percentile localization errors (P90 and P95), and Localiz ation Success Rate within radius r (LSR@r). Temporal experiments additionally report alon g-route Progress Mean Absolute Error (Progress MAE), Speed Mean Absolute Error (Speed M AE), abnormal jump statistics when available, end-to-end latency, and Frames Per Second (FP

S) when runtime throughput is reported. Lower MLE, MedLE, P90, P95, Progress MAE, and l atency are better; higher LSR is better.

5.2 MS and Weighted-Centroid Accuracy–Runtime Trade-off

This experiment isolates the spatial decoder by comparing probability-weighted centroid decoding with the final Mean-Shift (MS) aggregation under identical candidate responses. Tab le 2 summarizes both localization accuracy and the isolated coordinate-aggregation cost. In Ta ble 2, Avg. aggregated coordinates indicates how many candidate coordinates contribute to the final decoded point on average; MLE and P90 measure mean and tail error, LSR@r gives the percentage of frames localized within r meters, and Aggregation time measures only the final coordinate-combination step. The timing therefore excludes image preprocessing, backbone en coding, visual matching, iterative MS updates, GRU inference, and Kalman fusion.

Table 2. Accuracy and coordinate-aggregation runtime comparison between weighted centroid and MS. Weighted Centroid averages all 18 retained candidate centers using the normalized response weights. MS instead performs local mode seeking and combines the converged modes; Avg. aggregated coordinates summarizes how concentrated the final spatial aggregation becomes after decoding.

Decoder	Avg. aggregated co ordinates	MLE (m)	P90 (m)	LSR@3 (%)	LSR@5 (%)	LSR@10 (%)	LSR@15 (%)	Aggregation time (ms)
Weighted Centroid	18	3.546	6.953	47.963	**77.646**	98.189	**99.377**	0.02755
Mean-Shift (MS)	1.04	**3.482**	**6.620**	**49.208**	76.995	**98.359**	99.293	**0.02141**

As shown in Table **錯誤! 找不到參照來源。**, MS reduces MLE from 3.546 m to 3.482 m and P90 from 6.953 m to 6.620 m, while increasing LSR@3 from 47.963% to 49.208%. We ighted Centroid is slightly higher at LSR@5 and LSR@15, so the decoder change should be in terpreted as improving the mean and tail of the error distribution rather than producing a unifo rm gain at every threshold. Because the microbenchmark excludes the iterative MS update itse lf, the aggregation-time column alone is not a complete speed–accuracy comparison; Table **錯誤! 找不到參照來源。** provides the corresponding end-to-end online latency comparison.

5.3 FieldAnchor-LR Component Ablation

This ablation evaluates the contribution of the major FieldAnchor-LR components using the same held-out Routes B and C. Table 3 adds the components in the same order as the prop osed pipeline: MS only is the frame-level continuous visual measurement; MS + 3-frame GR U adds temporal representation; MS + GRU + inertial polynomial adds explicit velocity–acc eleration propagation; and the final row adds the learned-variance Kalman Filter. MLE, P90, a nd LSR measure localization accuracy, while Progress MAE measures the along-route state er

ror. Reading the rows cumulatively makes it possible to identify whether an individual module is useful by itself or only after it is fused with the subsequent estimator.

Table 3. Progressive ablation of the FieldAnchor-LR localization framework. The rows are cumulative rather than independent alternatives: each row retains the components above it and adds the newly named module. Progress MAE is reported in meters along the planned route, and lower values indicate a state that follows the prescribed route progress more closely.

Configuration	MLE	P90	LSR@3	LSR@5	LSR@10	LSR@15	Progress MAE
MS only	4.705	9.072	35.739	57.357	94.992	**99.321**	10.889
MS + 3-frame GRU	4.795	9.362	33.588	57.838	94.482	99.236	10.914
MS + GRU + inertial polynomial	4.818	9.502	34.946	57.810	94.284	99.293	10.782
FieldAnchor-LR (+ learned-variance KF)	**3.482**	**6.620**	**49.208**	**76.995**	**98.359**	99.293	**0.195**

As shown in Table 3, the largest improvement appears when the temporal motion state is fused with the learned-variance Kalman Filter. Relative to the polynomial-only row, the complete FieldAnchor-LR configuration reduces MLE from 4.818 m to 3.482 m, P90 from 9.502 m to 6.620 m, and Progress MAE from 10.782 m to 0.195 m. The intermediate rows therefore show that the GRU and inertial polynomial mainly provide a structured motion state whose benefit is realized by uncertainty-aware fusion with the visual measurement.

5.4 Hard Forward Candidate Restriction

This experiment evaluates the hard forward candidate restriction that defines the term "Hard" in FieldAnchor-LR and examines whether it can reduce online computation while maintaining localization accuracy. Table 4 compares the complete 6×6 local geometry, which scores all 36 candidates, with the hard causal forward 3×6 geometry, which scores only the 18 candidates in front of the current route direction. The Candidates column is therefore the number of online satellite patches that are projected and scored, MLE/P90/LSR describe localization accuracy, and Latency/FPS describe the end-to-end operating cost under otherwise unchanged model settings.

Table 4. Comparison of complete 6×6 and hard forward 3×6 candidate restriction. Complete 6×6 scores all 36 candidates in the local neighborhood. Hard forward 3×6 retains only the 18 candidates selected by the causal route-aligned projection; all remaining visual, temporal, and filtering settings are unchanged.

Search	Candidates	MLE	P90	LSR@3	LSR@5	LSR@10	LSR@15	Latency (ms)	FPS
Full 6 × 6	36	3.766	7.225	42.898	69.836	**98.727**	99.887	69.993	14.287
Forward 3 × 6 (FieldAnchor-LR)	18	**3.501**	**6.948**	**49.689**	**72.864**	98.642	**100.000**	**69.490**	**14.390**

As shown in Table 4, Hard forward 3×6 improves MLE from 3.766 m to 3.501 m and P90 from 7.225 m to 6.948 m while reducing the number of scored candidates from 36 to 18. LSR@3 and LSR@5 also increase, and the measured latency changes from 69.993 ms to 69.490 ms. LSR@10 is marginally higher for Complete 6×6, but both settings are essentially saturated at the relaxed 10–15 m thresholds, indicating that the rear half-grid contributes little useful localization evidence under the tested route-conditioned protocol.

5.5 Temporal Window Length

This experiment verifies how much temporal history is required by the recurrent model. Table **錯誤! 找不到參照來源。** compares one-, two-, and three-frame inputs under the same visual decoder and final estimator. The one-frame setting uses the current observation without explicit temporal differences; the two-frame setting exposes the most recent first difference; and the three-frame FieldAnchor-LR setting makes the temporal mean, first difference, and second difference available to the GRU. Progress MAE is included together with MLE, P90, and LSR to show whether the temporal representation improves not only point accuracy but also along-route state consistency.

Table 5. Effect of the temporal input window. The temporal input changes only the amount of recent visual history supplied to the recurrent model. The three-frame setting corresponds to the complete FieldAnchor-LR temporal representation described in Section 4.6.

Temporal input	MLE	P90	LSR@3	LSR@5	LSR@10	LSR@15	Progress MAE
1 frame	3.919	7.666	44.171	69.128	**98.359**	**99.434**	0.201
2 frames	3.936	7.428	44.001	67.968	98.132	99.179	0.306
3 frames (FieldAnchor-LR)	**3.482**	**6.620**	**49.208**	**76.995**	**98.359**	99.293	**0.195**

As shown in Table **錯誤! 找不到參照來源。**, the three-frame input obtains the lowest MLE (3.482 m), P90 (6.620 m), and Progress MAE (0.195 m), and it raises LSR@5 to 76.995%. The two-frame setting does not consistently improve over one frame, whereas three frames are the first configuration that exposes both a first temporal difference and a second temporal difference to the GRU. The result is therefore consistent with the temporal representation defined in Equations (11)–(13) and illustrated in Figure 7.

5.6 Second-Order Inertial Motion Model

The motion ablation isolates the effect of the acceleration term in the inertial prediction. Table **錯誤! 找不到參照來源。** compares GRU velocity-only propagation with the second-order velocity-plus-acceleration polynomial used by FieldAnchor-LR while keeping the visual l

ocalization, recurrent structure, and final evaluation protocol unchanged. In addition to MLE, P90, and LSR, Progress MAE measures along-route position consistency and Speed MAE me asures the error of the learned per-frame motion, so the table evaluates both localization accur acy and the quality of the propagated motion state.

Table 6. Ablation of the inertial motion prediction. The first row uses the GRU velocity output for propagation. The second row additionally uses the predicted acceleration through the second-order inertial polynomial; Speed MAE evaluates the accuracy of the resulting per-frame motion estimate.

Motion model	MLE	P90	LSR@3	LSR@5	LSR@10	LSR@15	Progress MAE	Speed M AE
GRU velocity only	3.690	7.567	46.972	73.543	**98.529**	**99.547**	0.229	0.794
GRU velocity + acceleration polyn omial (FieldAnchor-LR)	**3.482**	**6.620**	**49.208**	**76.995**	98.359	99.293	**0.195**	**0.746**

As shown in Table **錯誤! 找不到參照來源。**, adding the acceleration term reduces MLE from 3.690 m to 3.482 m, P90 from 7.567 m to 6.620 m, Progress MAE from 0.229 m to 0.19 5 m, and Speed MAE from 0.794 to 0.746 m per frame. The gains are concentrated in mean er ror, tail error, strict-threshold success, and motion consistency rather than the already saturated 10–15 m success region. This supports retaining the second-order velocity-plus-acceleration f orm instead of velocity-only propagation.

5.7 Kalman Filtering and Measurement Variance

This experiment isolates the final state-estimation stage using the configurations availabl e in the versioned FieldAnchor-LR results. Table 7 compares the recurrent visual measuremen t without Kalman fusion against the complete learned-variance Kalman Filter. Both rows use t he same upstream visual and temporal pipeline; the difference is whether the motion predictio n and visual measurement are fused using the learned measurement covariance. The comparis on therefore measures the contribution of uncertainty-aware Kalman fusion rather than changi ng the visual search condition.

Table 7. Effect of Kalman filtering and learned measurement variance. No Kalman Filter uses the recurrent visual measurement directly. The FieldAnchor-LR row fuses the same upstream measurement with the inertial prediction using the learned measurement covariance and the Kalman update defined in Equations (19)–(21).

Final estimator	MLE	P90	LSR@3	LSR@5	LSR@10	LSR@15
No Kalman Filter	4.818	9.502	34.946	57.810	94.284	**99.293**
Learned-variance KF (FieldAnch or-LR)	**3.482**	**6.620**	**49.208**	**76.995**	**98.359**	**99.293**

As shown in Table 7, the learned-variance Kalman Filter reduces MLE from 4.818 m to 3.482 m and P90 from 9.502 m to 6.620 m. LSR@3 rises from 34.946% to 49.208% and LSR@5 from 57.810% to 76.995%, whereas LSR@15 is unchanged because both configurations already place nearly all frames inside the broad 15 m radius. The filter therefore improves the precision of moderate-error frames through covariance-aware fusion of the visual measurement and inertial prediction.

5.8 Route-Wise Final Localization Results

The complete FieldAnchor-LR model is evaluated separately on held-out Routes B and C, and Table 8 then aggregates all held-out frames in the B + C row. The MLE and Median columns describe central localization error, P90 and P95 expose the tail of the error distribution, and LSR@3 through LSR@20 report the fraction of frames inside increasingly relaxed metric radii. Presenting the two routes separately is important because similar mean errors can still hide different median and tail behavior.

Table 8. Final FieldAnchor-LR localization results on held-out routes. Route B and Route C are reported independently, and B + C pools all 3,534 held-out frames. Lower MLE/Median/P90/P95 and higher LSR indicate better localization performance.

Route	MLE	Median	P90	P95	LSR@3	LSR@5	LSR@10	LSR@15	LSR@20
Route B	3.437	3.272	6.512	7.633	45.387	76.582	99.605	100.000	100.000
Route C	3.565	2.624	6.909	8.374	56.121	77.742	96.105	98.013	99.205
B+C (all frames)	3.482	3.070	6.620	7.847	49.208	76.995	98.359	99.293	99.717

As shown in Table **錯誤! 找不到參照來源。**, the complete model reaches an aggregate MLE of 3.482 m, a median error of 3.070 m, P90 of 6.620 m, and P95 of 7.847 m across the 3,534 held-out frames. Route B and Route C have similar mean errors (3.437 m and 3.565 m), but their distributions differ: Route C has the lower median and higher LSR@3, while Route B has the lighter tail and higher success rates at the relaxed 10–20 m thresholds. Reporting the routes separately therefore reveals behavior that would be hidden by the B+C average alone.

5.9 Comparison with Existing UAV–Satellite Localization Methods

The external comparison follows each method's native localization protocol when a versioned output is available. Table 錯誤! 找不到參照來源。 therefore includes a Native protocol column before the numerical metrics: DenseUAV [7], Sample4Geo [4], and Game4Loc [8] use fixed-gallery global retrieval, InfoGeo [10] uses global retrieval when a versioned output is available, Bearing-UAV [9] is listed as neighbor-map regression, and FieldAnchor-LR uses the controlled local-search protocol described in this thesis. MLE, Median, P90, P95, and LSR provide the common error scale, while FPS gives the measured throughput when available. An em dash indicates that a versioned official-protocol result is not available in the current result package; it is not treated as a zero or estimated value.

Table 9. Comparison with external UAV–satellite localization methods on FieldAnchor. The Native protocol column must be read together with the metrics because global retrieval, neighbor-map regression, and controlled local search have different search scopes. The table is therefore used for protocol-aware error-scale context rather than a strictly matched architecture ranking.

Method	MLE	Median	P90	P95	LSR@3	LSR@5	LSR@10	LSR@15	FPS
DenseUAV [7]	417.282	347.952	872.145	1073.156	0.255	0.623	1.019	1.868	99.677
Sample4Geo [4]	284.905	241.941	588.841	709.878	0.340	1.330	2.320	3.537	226.264
Game4Loc [8]	184.182	124.480	434.277	559.439	2.235	5.065	7.838	12.054	229.348
InfoGeo [10]	238.403	132.820	541.617	731.367	0.509	1.500	4.160	7.612	—†
Bearing-UAV [9]	215.277	164.577	461.801	542.675	0.057	0.113	0.340	0.990	98.569
FieldAnchor-LR (Ours)	3.482	3.070	6.620	7.847	49.208	76.995	98.359	99.293	15.137

As shown in Table 錯誤! 找不到參照來源。, the available fixed-gallery global-retrieval outputs have a substantially larger error scale on FieldAnchor than the controlled local-relocalization setting used by FieldAnchor-LR. This numerical gap must be read together with the Native protocol column: the methods do not operate under the same search scope, so Table 錯誤! 找不到參照來源。 is not used to claim a direct architecture-only ranking. InfoGeo and Bearing-UAV remain marked with an em dash until versioned outputs under their documented protocols are available.

In addition to the protocol-aware numerical comparison in Table 9, a qualitative cross-method comparison is reserved in Figure 10. The final version should reproduce outputs from the available external methods on the same FieldAnchor queries and visualize each method's predicted position together with the corresponding predefined route reference point and metric error. Because the compared methods retain their native search protocols, this figure is intended to expose characteristic success and failure cases rather than to imply a strictly matched architecture ranking.

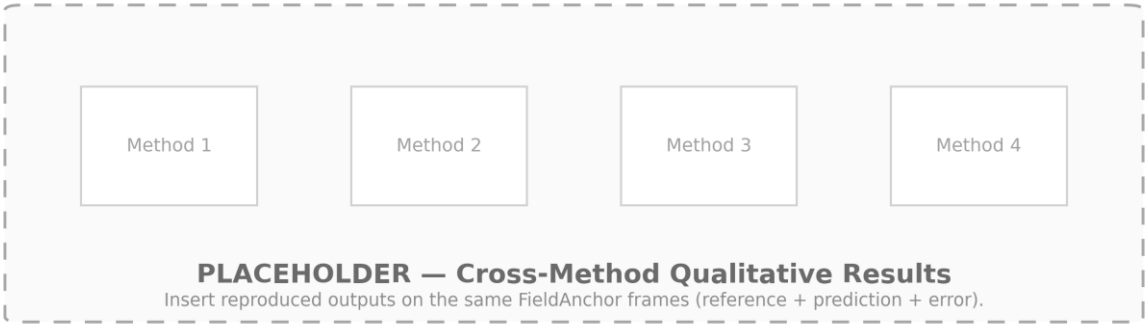

Figure 10. Cross-method qualitative localization results on FieldAnchor.

For the final submission, each panel in Figure 10 should use the same held-out UAV frame and map region for all reproduced methods. A compact and interpretable layout is to show the UAV query, the method prediction on the orthomosaic, the predefined route reference point, and the localization error in meters. This keeps the qualitative comparison aligned with Table 9 while avoiding an unfair visual comparison caused by using different query frames or map extents.

5.10 Backbone Accuracy–Runtime Trade-off on NVIDIA Jetson Orin NX

To evaluate the deployment cost of the visual encoder, four backbones are benchmarked on an NVIDIA Jetson Orin NX development platform under the same backbone-comparison p rocedure. MobileNetV3-Small, ResNet-18, VGG16, and MobileCLIP2-S2 are evaluated on th e held-out B+C split. End-to-end (E2E) average latency and P95 latency report complete per-f rame inference time for this benchmark on the Orin NX platform, approximate FPS summariz es throughput, and B+C MLE measures localization accuracy. The current benchmark record d oes not specify the power mode, numerical precision, or acceleration backend, so those setting s are not inferred here.

Table 10. Backbone accuracy–latency trade-off on the NVIDIA Jetson Orin NX development platform. All latency and throughput values in Table **錯誤! 找不到參照來源。** are measured on the same NVIDIA Jetson Orin NX development platform. Lower E2E latency, P95 latency, and MLE are better, whereas higher FPS is better. The B+C MLE values belong specifically to this backbone deployment sweep and should therefore be interpreted within this comparison rather than substituted for the final aggregate result reported in Table 8.

Backbone	E2E mean latency	P95 latency	Approx. FPS	B+C MLE
MobileNetV3-Small	88.83 ms	91.60 ms	11.26	4.224 m
ResNet-18	95.07 ms	97.85 ms	10.52	4.077 m
VGG16	137.35 ms	147.53 ms	7.28	4.277 m
MobileCLIP2-S2	188.55 ms	191.44 ms	5.30	3.634 m

As shown in Table **錯誤! 找不到參照來源。**, the backbone benchmark exhibits a clear a ccuracy–throughput trade-off on the Jetson Orin NX. MobileNetV3-Small is the fastest config uration, with 88.83 ms average E2E latency, 91.60 ms P95 latency, and approximately 11.26 F PS, but its B+C MLE is 4.224 m. ResNet-18 retains similar throughput at 10.52 FPS while red ucing MLE to 4.077 m. VGG16 is slower and, in this experiment, also has the highest MLE a mong the four backbones, with 137.35 ms average latency and 4.277 m MLE. MobileCLIP2-S 2 provides the lowest B+C MLE of 3.634 m, but it also has the largest runtime cost, reaching 1 88.55 ms average latency, 191.44 ms at P95, and approximately 5.30 FPS. Relative to Mobile NetV3-Small, MobileCLIP2-S2 reduces MLE by 0.590 m (approximately 14.0%) while requir ing about 2.12× the average latency. These results indicate that the visual backbone should be selected as an accuracy–runtime trade-off for embedded deployment: lightweight CNNs offer higher on-device throughput, whereas MobileCLIP2-S2 provides the strongest localization acc uracy in the measured Orin NX backbone sweep.

5.11 Qualitative Trajectory Analysis

Two qualitative views are used to complement the scalar metrics. Figure 11 is reserved fo r a contiguous held-out sequence that exposes frame-to-frame temporal stability, while Figure 12 overlays the final filtered route on the FieldAnchor orthomosaic to show route-level contin uity.

Figure 11 should visualize one uninterrupted sequence rather than isolated frames. The re commended plot places frame index on the horizontal axis and either localization error or alon g-route progress on the vertical axis, with curves for MS-only, MS + three-frame GRU, MS + GRU + second-order inertia, and the complete FieldAnchor-LR estimator. This directly shows whether the recurrent and filtering components reduce jitter, overshoot, and abrupt one-frame corrections that may not be visible from MLE or P90 alone.

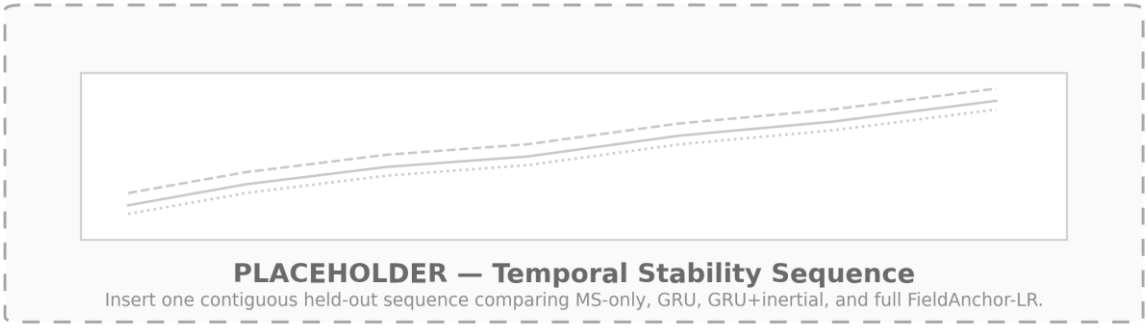

Figure 11. Temporal stability on a contiguous held-out FieldAnchor sequence. The final Figure 11 should use exactly the same frame interval for every configuration and should not smooth the plotted outputs for visualization. If along-route progress is used, the registered route progress can be included as a reference curve; if localization error is used, the plot should retain the original per-frame values so that transient spikes remain visible. The purpose is to demonstrate temporal behavior, not to duplicate the aggregate values already reported in Tables 3, 5, 6, and 7.

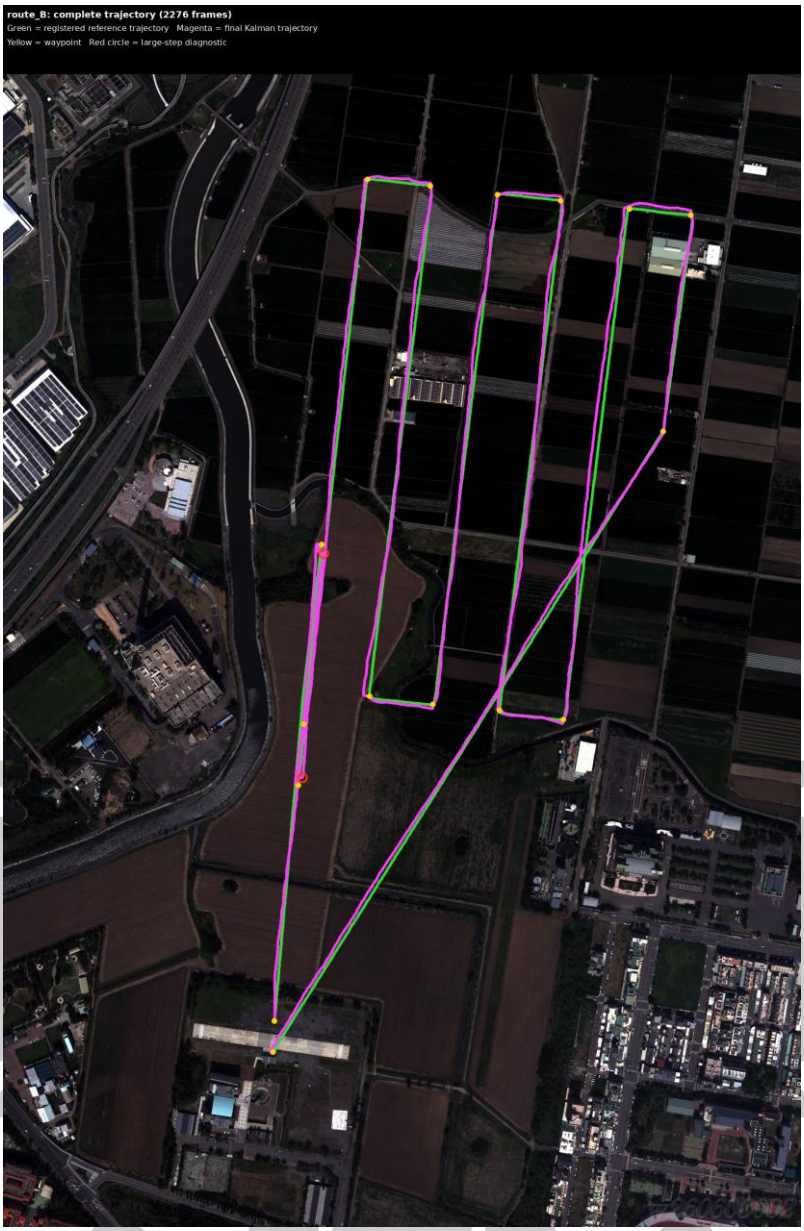

Figure 12. Final FieldAnchor-LR trajectory on the FieldAnchor orthomosaic. Figure 12 overlays the planned reference route and the final filtered trajectory to provide a route-level view of continuity that is not captured by scalar error statistics alone. The filtered path should be interpreted together with Figure 11: Figure 11 exposes local frame-to-frame stability, whereas Figure 12 shows whether those local updates remain coherent over the complete held-out route. Detailed quantitative conclusions remain in Tables 3–9.

Along straight route segments, the filtered estimate changes smoothly because the inertial prediction supplies a causal motion prior and the Mean-Shift observation supplies an uncertai nty-quantified visual measurement. This behavior is important in repetitive farmland, where se veral neighboring satellite crops can receive similar visual responses even when their metric c enters differ.

Near route turns, the recurrent model adapts the local motion trend through the estimated heading, velocity, and acceleration states, while the learned measurement covariance adjusts th e relative contribution of the current visual observation in the Kalman gain. The temporal mod el and uncertainty-aware filter therefore address motion continuity and visual ambiguity witho ut changing the definition of “Hard,” which remains exclusively the forward 3×6 candidate res triction.

The trajectory visualization is consequently used only for qualitative continuity analysis. Quantitative conclusions about component contributions, search geometry, temporal window l ength, motion prediction, Kalman fusion, and route-wise accuracy are reported once in Tables **錯誤! 找不到參照來源。–錯誤! 找不到參照來源。** and discussed directly beside those tabl es, avoiding duplicate bar charts or curves that repeat the same numerical evidence.

Chapter 6 Conclusion

This thesis presents FieldAnchor and the FieldAnchor Local Relocalization (FieldAnchor -LR) framework for route-conditioned UAV–satellite visual relocalization in repetitive farmland. The complete method combines a frozen MobileCLIP2-S2 representation, a controlled local-search condition, causal forward 3×6 candidate selection, continuous Mean-Shift (MS) visual localization, three-frame Gated Recurrent Unit (GRU) temporal modeling, a second-order inertial polynomial, learned visual-measurement variance, and an external uncertainty-aware Kalman Filter (KF) in continuous route coordinates. The “Hard” wording in the fixed thesis title refers exclusively to the explicit forward candidate restriction that retains the route-aligned 3×6 subset from the complete 6×6 local geometry before Mean-Shift decoding; it does not denote a separate MS variant.

On the held-out Routes B and C, the complete FieldAnchor-LR configuration achieves an overall Mean Localization Error of 3.482 m, a median error of 3.070 m, and a P90 of 6.620 m. The combined Localization Success Rates are 76.995% within 5 m, 98.359% within 10 m, and 99.293% within 15 m. The ablation results show that the learned-variance Kalman Filter provides the largest improvement in final localization accuracy, while the three-frame representation and second-order motion model provide the temporal state used by the final fusion stage. The hard forward 3×6 candidate restriction halves the number of online satellite candidates from 36 to 18 before visual scoring and Mean-Shift decoding, while the subsequent MS, GRU, inertial, and Kalman stages operate as separate components of the localization pipeline. The reported experiment is a route-conditioned local relocalization study around predefined planned-route reference points and does not claim unrestricted global acquisition from an unknown map position.

References

[1] S. Hu, M. Feng, R. M. H. Nguyen, and G. H. Lee, “CVM-Net: Cross-View Matching Network for Image-Based Ground-to-Aerial Geo-Localization,” Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 7258–7267, 2018.

[2] Y. Shi, L. Liu, X. Yu, and H. Li, “Spatial-Aware Feature Aggregation for Image-Based Cross-View Geo-Localization,” Advances in Neural Information Processing Systems, vol. 32, 2019.

[3] S. Zhu, M. Shah, and C. Chen, “TransGeo: Transformer Is All You Need for Cross-View Image Geo-Localization,” Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 1162–1171, 2022.

[4] F. Deuser, K. Habel, and N. Oswald, “Sample4Geo: Hard Negative Sampling for Cross-View Geo-Localisation,” Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 16847–16856, 2023.

[5] Z. Zheng, Y. Wei, and Y. Yang, “University-1652: A Multi-View Multi-Source Benchmark for Drone-Based Geo-Localization,” Proceedings of the 28th ACM International Conference on Multimedia, pp. 1395–1403, 2020.

[6] R. Zhu, L. Yin, M. Yang, F. Wu, Y. Yang, and W. Hu, “SUES-200: A Multi-Height Multi-Scene Cross-View Image Benchmark Across Drone and Satellite,” IEEE Transactions on Circuits and Systems for Video Technology, vol. 33, no. 9, pp. 4825–4839, 2023.

[7] M. Dai, E. Zheng, Z. Feng, L. Qi, J. Zhuang, and W. Yang, “Vision-Based UAV Self-Positioning in Low-Altitude Urban Environments,” IEEE Transactions on Image Processing, vol. 33, pp. 493–508, 2024.

[8] Y. Ji, B. He, Z. Tan, and L. Wu, “Game4Loc: A UAV Geo-Localization Benchmark from Game Data,” Proceedings of the AAAI Conference on Artificial Intelligence, vol. 39, no. 4, pp. 3913–3921, 2025.

[9] K. Liu, H. Zhou, R. Xu, P. Wang, M. Song, and H. Zhang, “Beyond Matching to Tiles: Bridging Unaligned Aerial and Satellite Views for Vision-Only UAV Navigation,” Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 5359–5368, 2026.

[10] H. Zhang, M. Wang, Z. Wang, H. Yin, and M. O. Pun, “InfoGeo: Information-Theoretic Object-Centric Learning for Cross-View Generalizable UAV Geo-Localization,” Proceedings of the 43rd International Conference on Machine Learning, PMLR 306, 2026.

[11] D. Comaniciu and P. Meer, “Mean Shift: A Robust Approach Toward Feature Space Analysis,” IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 24, no. 5, pp. 603–619, 2002.

[12] F. Faghri, P. K. A. Vasu, C. Koc, V. Shankar, A. Toshev, O. Tuzel, and H. Pouransari, “MobileCLIP2: Improving Multi-Modal Reinforced Training,” Transactions on Machine Learning Research, 2025.

[13] K. Cho, B. van Merrienboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, “Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation,” Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing, pp. 1724–1734, 2014.

[14] R. E. Kalman, “A New Approach to Linear Filtering and Prediction Problems,” Journal of Basic Engineering, vol. 82, no. 1, pp. 35–45, 1960.

[15] J.-W. Hsieh, C.-H. Chou, M.-C. Chang, P.-Y. Chen, S. Santra, and C.-S. Huang, “Mean-Shift Based Differentiable Architecture Search,” IEEE Transactions on Artificial Intelligence, vol. 5, no. 3, pp. 1235–1246, Mar. 2024, doi: 10.1109/TAI.2023.3329792. DOI

[16] Z. Li, X. Zhang, Y. Guo, Y. Xie, Z. Ding, S. Cai, H. Li, and M. Lao, “PAUL: Uncertainty-Guided Partition and Augmentation for Robust Cross-View Geo-Localization under Noisy Correspondence,” Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 5389–5398, 2026.

[17] X. Liang, H. Tang, F. Zhang, S. Yuan, C. Hu, D. Zheng, and K. Ma, “UniGeoRS: A Unified Benchmark for Tri-view Geo-Localization,” Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 5399–5408, 2026.

[18] Z. Song, J. Zhang, D. Wang, Z. Zhou, W. Liu, H. Guo, E. Wang, and B. Du, “GeoBridge: A Semantic-Anchored Multi-View Foundation Model Bridging Images and Text for Geo-Localization,” Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 27793–27803, 2026.

[19] X. Cheng, L. Wang, Y. Liu, X. Liu, H. Tan, Y. Liu, M. Zhang, and S. Yan, “PiLoT: Neural Pixel-to-3D Registration for UAV-based Ego and Target Geo-localization,” Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 5379–5388, 2026.

[20] Y. Zhang, X. Zhang, G. Sun, Z. Lyu, S. Wshah, and C. Chen, “Geo2: Geometry-Guided Cross-view Geo-Localization and Image Synthesis,” Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 19432–19442, 2026.